
The Collapse Model: Label-Free Data-Pool Compression via Spectral Geometry

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Abstract

Given M candidate datasets, which $k \ll M$ subset best preserves representational diversity when their features are merged—without access to labels or downstream tasks? We address this *label-free data-pool compression* problem by deriving the **Collapse Model**, a closed-form bridge between spectral geometry and compression decisions. The model predicts the merged effective rank (r_{eff}) from a 2×2 Gram-block eigenstructure via two compression-relevant SVD channels (energy ratio γ , directional alignment α), yielding an explicit superadditivity threshold $\rho_c(\alpha, \gamma)$ that drives a greedy compression algorithm using only *per-dataset summary statistics*—suitable for privacy-constrained or streaming settings where features cannot be shared. Across 427 real feature-matrix pairs from 5 encoder families and 2 modalities the formula attains Pearson $r=0.97$. Collapse-guided compression is near-optimal ($\leq 1\%$ regret vs. exhaustive search), matches the closest published spectral-diversity baseline (Vendi-Greedy) on retention while requiring only summary statistics rather than full feature matrices, and strictly outperforms Facility Location / k -Medoids on downstream linear-probe accuracy (sign test $p < 10^{-3}$, Collapse exceeds Random’s 95% CI on the mean LP). The comparison reveals two distinct LP-relevant diversity axes (*spectral retention* captured by Collapse and *centroid coverage* captured by farthest-first traversal), clarifying when each signal dominates. The framework scales to $M=43$ pools at up to $14\times$ compression and identifies a quantitative supervised / self-supervised spectral-geometry divide via a bimodal principal-angle distribution exposed in CLIP and DINO encoders. Code is available at <https://github.com/anonymization26/collapsemodel>.

1 Introduction

Modern pretrained models increasingly rely on combining features from many candidate datasets—multi-domain transfer learning, data curation for self-supervised pretraining [Gadre et al., 2023], and continual learning all face the question: *given M candidate datasets, which $k \ll M$ subset preserves the diversity of the full pool when their features are concatenated?* We call this the *label-free data-pool compression* problem: unlike classical coresets selection [Mirzasoleiman et al., 2020] and recent *sample-level* data curation methods (DSIR [Xie et al., 2023], DsDm [Engstrom et al., 2024], D² Pruning [Maharana et al., 2023]), we seek compression at the *dataset level* (not the sample level), without access to labels or downstream tasks. The criterion is representational diversity, formalized as the effective rank $r_{\text{eff}}(H) = \exp(-\sum_i p_i \log p_i)$ of the merged features [Roy and Vetterli, 2007, Garrido et al., 2023]. No prior work provides a closed-form answer to how r_{eff} changes under feature concatenation, forcing practitioners to rely on exhaustive search or heuristics that scale poorly with M .

Prior work addresses related but distinct problems. *Spectral diversity metrics* such as RankMe [Garrido et al., 2023], VNE [Kim et al., 2023], and the Vendi Score [Friedman and Dieng, 2023, Pasarkar and Dieng, 2024] quantify the diversity of a single set via entropy of a singular-value or similarity-matrix spectrum; they do not predict how two sets’ diversity combines under concatenation. *Label-free coreset selection* (ELFS [Shen et al., 2025], ZCore [Griffin et al., 2024], DEFT-UCS [Das and Khetan, 2024]) selects a subset of *samples* using geometry or proxy dynamics; it is complementary but orthogonal to our dataset-level formulation. *Pool-level curation* (DataComp [Gadre et al., 2023], DataComp-LM [Li et al., 2024], DsDm [Engstrom et al., 2024], Quad [Zhang et al., 2025]) uses filtering or model-aware heuristics without a closed-form diversity criterion. *Representation similarity measures* (SVCCA [Raghu et al., 2017], PWCCA [Morcos et al., 2018], CKA [Kornblith et al., 2019], PARC [Bolya et al., 2021]) compare two representations but do not predict merged rank. *Transferability estimators* (LogME [Zhang et al., 2021], H-score [Bao et al., 2019], TransRate [Huang et al., 2022], LEEP [Li et al., 2021], reassessed by Ibrahim et al. [2022]) and PMI-based dataset valuation [Zhang et al., 2024] predict downstream accuracy but either require labels or do not model merging. Task2Vec [Achille et al., 2019] avoids labels but requires probe fine-tuning. Model-merging methods (Task Arithmetic [Ilharco et al., 2023], TIES-Merging [Yadav et al., 2023], DARE [Yu et al., 2024]) combine parameters, not features. Intrinsic-dimensionality work [Pope et al., 2021, Ansuini et al., 2019] characterizes individual matrices but not merged ones. To our knowledge, no existing work provides either a closed-form prediction of $r_{\text{eff}}([H_A; H_B])$ or a theoretical framework that explains *when and why* feature merging increases diversity.

We derive the **Collapse Model**, a closed-form bridge between spectral geometry and label-free compression. It expresses the merged effective rank as a function of four per-dataset quantities— r_A , r_B , the energy ratio $\gamma = \|H_B\|_* / \|H_A\|_*$, and the subspace alignment $\alpha = \text{SA}_k(H_A, H_B)$ —via a 2×2 Gram block eigenstructure. Lemma 1 gives exact eigenvalues under the principal-angle model (discriminant D , weight r^*); Proposition 2 approximates the merged spectrum as a two-component mixture with closed-form entropy. The resulting formula identifies a superadditivity threshold $\rho_c(\alpha, \gamma)$ and partitions the (α, γ) plane into **three behaviour regimes** (Corollary 3): *superadditive* (balanced energy and low alignment, $r_{\text{dom}}/r_{\text{sub}} < \rho_c$), *directional collapse* ($\alpha \rightarrow 1$, subspaces overlap so merging adds no diversity), and *energy collapse* ($\gamma \gg 1$ or $\gamma \ll 1$, one dataset dominates and the gain vanishes). Our SA_k uses mean squared cosine of principal angles, which connects more directly to projection theory than SVCCA’s unsigned cosine or PWCCA’s energy-weighted form; the U -channel uses the Two-NN intrinsic dimension estimator [Facco et al., 2017].

Why closed-form compression? Exhaustive search over M candidates requires $\binom{M}{k}$ merged SVDs at $O(Nd^2)$ each, with simultaneous access to all feature matrices ($M=14, k=5$: 2002 SVDs; $M=100, k=10$: 1.7×10^{13}). The Collapse Model reduces this to $O(kM)$ rank predictions from per-dataset summary statistics, enabling compression in privacy-constrained and streaming settings.

Contributions. (1) *Problem formulation—label-free data-pool compression* (Sec. 5): an $\mathcal{O}(kM)$ greedy algorithm operating from per-dataset summary statistics ($r_{\text{eff}}, \|\cdot\|_*, V_{\text{top-}k}$), suitable for privacy-constrained and streaming settings where feature matrices cannot be shared. (2) *Closed-form theory* (Lemma 1, Proposition 2, Theorem 1): the first closed-form predictor of merged r_{eff} , decomposing it into three orthogonal SVD channels with a superadditivity threshold $\rho_c(\alpha, \gamma)$ (Corollary 3) that classifies merging into three regimes (superadditive / directional collapse / energy collapse) and an explicit multiplicative error bound. (3) *Two-signal decomposition of LP-relevant diversity* (Sec. 5): benchmarking against modern baselines reveals that downstream-relevant diversity factors into two orthogonal signals—*spectral retention* (captured by Collapse, $r_A + r_B$, Vendi-Greedy) and *centroid coverage* (captured by farthest-first traversal). A two-stage Cen→Col hybrid composes both signals and dominates pure Collapse on *both* LP and retention simultaneously at $k=10$ (Appendix J), positioning the closed-form formula as a composable primitive rather than a standalone strategy.

Empirical findings. The formula is validated on 280 controlled + 120 non-principal-angle synthetic + 427 real pairs across 5 encoder families and 2 modalities (Pearson $r=0.97$, $R_{\text{cal}}^2=0.95$). The framework identifies a quantitative supervised/self-supervised spectral-geometry divide: ResNet-50 achieves 100% superadditivity across 91 pairs, while CLIP and DINO reach only 29–36% on cross-domain merges—a gap mechanistically explained by a bimodal principal-angle distribution that the model isolates as the responsible mechanism.

Among label-free analyses, only the Collapse Model simultaneously provides closed-form merged-rank prediction ($R_{\text{cal}}^2=0.95$ on 427 pairs, Table 2), behaviour-regime classification via $\rho_c(\alpha, \gamma)$,

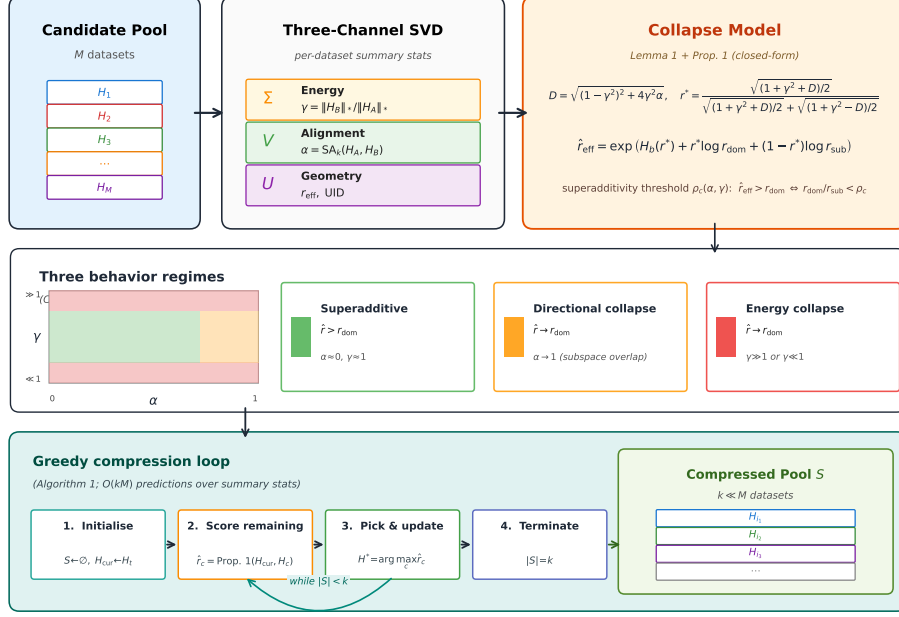


Figure 1: Label-free data-pool compression pipeline. **Top:** Three SVD-channel statistics (Σ : γ , V : α , U : r_{eff}) per dataset feed Lemma 1+Prop. 2 to predict merged rank and superadditivity threshold $\rho_c(\alpha, \gamma)$. **Middle:** Three behaviour regimes in the (α, γ) plane (Cor. 3): superadditive, directional collapse ($\alpha \rightarrow 1$), energy collapse ($\gamma \gg 1$ or $\ll 1$). **Bottom:** Greedy loop (Alg. 1): score remaining candidates, pick arg max, repeat k times; outputs compressed pool S .

92 and superadditivity diagnosis at the same $O(Nd \min(N, d))$ cost as SVCCA/PWCCA; the naive
 93 $r_A + r_B$ matches Collapse in ordinal ranking ($\rho=0.963$ vs. 0.966, not significant) but lacks absolute
 94 R^2 , regime diagnosis, and redundancy detection.

95 2 Background: SVD Three-Channel Decomposition

96 For any feature matrix $H \in \mathbb{R}^{N \times d}$ with thin SVD $H = U\Sigma V^\top$ ($U \in \mathbb{R}^{N \times r}$, $\Sigma = \text{diag}(\sigma_1 \geq \dots \geq$
 97 $\sigma_r > 0)$, $V \in \mathbb{R}^{d \times r}$), the decomposition separates three orthogonal aspects of representation.

98 **Σ channel — spectral energy.** The **effective rank** summarizes spectral distribution:

$$r_{\text{eff}}(H) = \exp\left(-\sum_{i=1}^r p_i \log p_i\right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}, \quad (1)$$

99 where p_i is the singular-value share (not squared), so $r_{\text{eff}} \in [1, r]$. The nuclear norm $\|H\|_* = \sum_i \sigma_i$
 100 captures total energy. A spectrum-difference quantity (Spectral Transfer Gap, used in the Gram-block
 101 transfer bound of Appendix C) is defined there.

102 **V channel — directional structure.** Right singular vectors V encode which directions are activated.

103 **Definition 1** (Subspace Alignment). Let $\hat{V}_A^k, \hat{V}_B^k \in \mathbb{R}^{d \times k}$ be the top- k right singular vectors of
 104 H_A, H_B . The **Subspace Alignment** at level k is:

$$\text{SA}_k(H_A, H_B) = \frac{1}{k} \left\| \hat{V}_A^{k\top} \hat{V}_B^k \right\|_F^2 = \frac{1}{k} \sum_{i=1}^k \cos^2 \theta_i, \quad (2)$$

105 where $\theta_1 \leq \dots \leq \theta_k$ are the principal angles.

106 $\text{SA}_k \in [0, 1]$: $\text{SA}_k = 0$ means perfectly orthogonal subspaces (maximum complementarity); $\text{SA}_k =$
 107 1 means identical subspaces (no new directions from merging).

108 **U channel — sample geometry.** Left singular vectors $U \in \mathbb{R}^{N \times r}$ encode the geometric distribution
 109 of samples. The **U-space Intrinsic Dimension (UID)** via the Two-NN estimator:

$$\text{UID}(H) = \frac{1}{\log 2 \cdot \mathbb{E}[\log(\mu_2/\mu_1)]}, \quad (3)$$

110 where μ_1, μ_2 are geodesic distances to the nearest and second-nearest neighbors on the unit sphere
 111 S^{k-1} (L2-normalized U_k rows).

112 **Role of each channel.** γ and α enter the merged-rank prediction formula (Sec. 3) and drive the
 113 compression algorithm. UID characterizes the intrinsic geometry of individual datasets and is used
 114 for encoder diagnostics (Sec. 4.2) but does *not* appear in the prediction formula; it is included here to
 115 complete the orthogonal decomposition of representation space.

116 **Independence.** Pairwise Spearman tests (Bonferroni-corrected, $n=15$ per probe) detect no significant
 117 correlation among the three channels on ResNet-50 or ViT-B/16 (all $p_{\text{Bonf}} > 0.38$), consistent with
 118 γ, α , UID capturing distinct properties.

119 3 The Collapse Model

120 3.1 Setup and Notation

121 Let $H_A \in \mathbb{R}^{N_A \times d}$ and $H_B \in \mathbb{R}^{N_B \times d}$ be L2-row-normalized feature matrices with effective ranks
 122 $r_A = r_{\text{eff}}(H_A)$, $r_B = r_{\text{eff}}(H_B)$, nuclear norms $\|H_A\|_*$, $\|H_B\|_*$, and subspace alignment $\alpha =$
 123 $\text{SA}_k(H_A, H_B)$. Define:

$$\gamma = \frac{\|H_B\|_*}{\|H_A\|_*} \in (0, \infty), \quad r_{\text{dom}} = \begin{cases} r_A & \gamma \leq 1 \\ r_B & \gamma > 1 \end{cases}, \quad r_{\text{sub}} = \begin{cases} r_B & \gamma \leq 1 \\ r_A & \gamma > 1 \end{cases}. \quad (4)$$

124 γ is the **energy ratio** of the two matrices; r_{dom} is the effective rank of the higher-energy matrix.

125 3.2 Main Results

126 Our derivation has two layers: an *exact* algebraic result (Lemma 1) and an *approximate* prediction
 127 formula (Proposition 2) that applies the exact result via a spectral mixture model.

128 **Lemma 1** (Exact: 2×2 Gram Block Eigenstructure). *Under the principal-angle model where the i -th*
 129 *right singular vector of H_B lies in $\text{span}\{V_{A,i}, V_{\perp,i}\}$ with $V_{A,i}^\top V_{B,i} = \cos \theta_i$, the 2×2 block of the*
 130 *Gram matrix $G = H_A^\top H_A + H_B^\top H_B$ in the $(V_{A,i}, V_{\perp,i})$ basis has eigenvalues $\sigma_{A,i}^2 (A \pm D_i)/2$ with*
 131 *$A = 1 + \gamma^2$ and*

$$D = \sqrt{(1 + \gamma^2)^2 - 4\gamma^2(1 - \alpha)} = \sqrt{(1 - \gamma^2)^2 + 4\gamma^2 \alpha}, \quad (5)$$

132 where $\alpha = \cos^2 \theta$ is the (shared) squared alignment. The nuclear-norm fraction of the dominant
 133 merged component is

$$r^* = \frac{\sqrt{(A + D)/2}}{\sqrt{(A + D)/2} + \sqrt{(A - D)/2}}. \quad (6)$$

134 This result is exact under the stated model (proof: Appendix B, Part A).

135 **Proposition 2** (Collapse Model: Predicted Merged Rank). *Under the two-component spectral mixture*
 136 *approximation (Appendix B, Part B), the predicted merged effective rank is:*

$$\hat{r}_{\text{merged}} = \exp\left(H_b(r^*) + r^* \log r_{\text{dom}} + (1 - r^*) \log r_{\text{sub}}\right), \quad (7)$$

137 where r^* is from Lemma 1, $H_b(p) = -p \log p - (1 - p) \log(1 - p)$ is binary entropy, and $r_{\text{dom}}, r_{\text{sub}}$
 138 are the effective ranks of the higher- and lower-energy matrices. The approximation error is
 139 $O(\sigma_{\min}/\sigma_{\max})$, controlled by γ .

140 **Interpretation.** Equation (7) is a weighted log-geometric mean of the individual ranks plus a balance
 141 bonus $H_b(r^*)$ (maximal when $r^*=0.5$, vanishing as $r^* \rightarrow 0, 1$).

142 **Behavior regimes.** *Superadditive* ($\alpha \approx 0, \gamma \approx 1$): $r^* \approx 0.5, \hat{r}_{\text{merged}} \approx 2\sqrt{r_A r_B} \geq r_{\text{dom}}$. *Directional*
 143 *collapse* ($\alpha \rightarrow 1$): $r^* \rightarrow 1, H_b(r^*) \rightarrow 0, \hat{r}_{\text{merged}} \rightarrow r_{\text{dom}}$. *Energy collapse* ($\gamma \gg 1$ or $\gamma \ll 1$): $r^* \rightarrow 1,$
 144 $\hat{r}_{\text{merged}} \rightarrow r_{\text{dom}}$. When $\alpha \rightarrow 1$ and $\gamma \gg 1$, the merged rank can fall below r_{sub} —a net loss in diversity.

145 3.3 Superadditivity Threshold

146 **Corollary 3** (Bounded-Ratio Superadditivity). *Proposition 2 predicts $\hat{r}_{\text{merged}} > r_{\text{dom}}$ if and only if*

$$\frac{r_{\text{dom}}}{r_{\text{sub}}} < \rho_c(\alpha, \gamma) \equiv \exp\left(\frac{H_b(r^*)}{1 - r^*}\right), \quad (8)$$

147 *where r^* is from Lemma 1. The threshold ρ_c is increasing in α : $\rho_c \geq 4.0$ for all $\alpha \in [0, 0.9]$ at*
 148 *$\gamma = 1$. At $\alpha = 1$: $H_b(r^*) \rightarrow 0$, $\rho_c \rightarrow 1$ —superadditivity is impossible.*

149 **Physical meaning.** ρ_c is a rank-ratio threshold: orthogonal subspaces ($\alpha \rightarrow 0$) allow $\rho_c \gg 1$ (very
 150 unbalanced pairs remain superadditive); identical subspaces ($\alpha \rightarrow 1$) force $\rho_c \rightarrow 1$ (no gain). The
 151 two-part proof and principal-angle scope are in Appendix B, N.

152 3.4 Approximation Error Bound

153 The two-component spectral mixture in Proposition 2 replaces the per-dimension discriminants D_i
 154 with a single value $\bar{D}$ computed from the average alignment $\bar{\alpha} = \text{SA}_k$. Theorem 1 formalizes when
 155 this approximation is exact and how its error depends on observable quantities.

156 **Theorem 1** (Spectral Mixture Error Bound). *Under the principal-angle model with per-dimension*
 157 *alignments $\alpha_i = \cos^2 \theta_i$, average alignment $\bar{\alpha} = \text{SA}_k$, energy ratio γ , and condition number*
 158 *$\kappa = \sigma_1^A / \sigma_k^A$, let $\hat{r}$ denote the Collapse Model prediction (Proposition 2) and $r = r_{\text{eff}}([H_A; H_B])$ the*
 159 *true merged effective rank.*

160 (a) **Exactness under uniform alignment.** *If $\alpha_i = \bar{\alpha}$ for all $i = 1, \dots, k$, then $\hat{r} = r$.*

161 (b) **General error bound.**

$$\left| \log \frac{\hat{r}}{r} \right| \leq \frac{4\gamma^2}{\bar{D}^2} \cdot V_\alpha \cdot \log \kappa, \quad (9)$$

162 *where $\bar{D} = \sqrt{(1 - \gamma^2)^2 + 4\gamma^2 \bar{\alpha}}$ is the average discriminant and $V_\alpha = \sum_{i=1}^k p_i (\alpha_i - \bar{\alpha})^2$ is the*
 163 *spectrum-weighted alignment variance with $p_i = \sigma_{A,i} / \|H_A\|_*$.*

164 The three factors capture discriminant sensitivity, alignment heterogeneity, and spectral non-
 165 uniformity respectively, and combine to the worst-case guarantee $|\log(\hat{r}/r)| \leq \log \kappa$ (multiplicative
 166 error never exceeds the condition number). Full proof and the discussion of why the formula
 167 generalises beyond the principal-angle model (E1a-Ind, $r=0.9987$) are in Appendices D, N.

168 **Calibration.** On real data the formula acts as a systematic upper bound; a single multiplicative con-
 169 stant c restores quantitative accuracy and follows a universal regression $c=0.946-0.175 \log(N/r_{\text{eff}})$
 170 across encoder families (Appendix N). Because this is a *one-parameter* linear model in the observable
 171 $\log(N/r_{\text{eff}})$, a reliable estimate of c requires only 5–10 labeled pairs per encoder family—a negligible
 172 overhead compared to the cost of computing frozen-encoder features. In privacy-constrained settings
 173 where even summary statistics from a single reference pair are available, the per-encoder c can be
 174 estimated without exposing any raw features. The same three channels also bound linear-probe
 175 transfer loss (Appendix C): low SA_k implies *high* merge gain and *high* transfer loss simultaneously.

176 4 Formula Validation: Closing the Theoretical Loop

177 Before applying the Collapse Model to compression (Sec. 5), we establish a three-step validation
 178 chain: (i) **mathematical correctness**—the formula is exact under its derivation assumptions (§4.1,
 179 synthetic grid); (ii) **real-data generalization**—the formula remains a strong approximation across 5
 180 encoder families, 14 vision datasets, and 19 text corpora (§4.2, 427 pairs); (iii) **behavioral regime**
 181 **agreement**—empirical superadditivity patterns match the three regimes predicted by Corollary 3
 182 (§4.2, Table 2). Together these close the theoretical loop: the formula is correct, it generalizes, and its
 183 qualitative predictions are observable. Additional controls (E1a-Ind 120 non-principal-angle pairs,
 184 E1b 30 real pairs, E1c 57 pairs, E_diverse cross-architecture, E_downstream r_{eff} /LP-accuracy) all
 185 agree with the same picture and are deferred to Appendix O. Feature matrices are L2-row-normalized
 186 and subsampled to N_{sub} samples; effective rank uses Eq. (1); bootstrap CIs use $B = 10,000$
 187 resamples.

Table 1: E1a: Synthetic grid results ($N = 280$). Naive baseline fails at large γ .

Method	R^2	Max error	Superadd. (%)	$\alpha < 1$ superadd.
Naive $\sqrt{1 + \alpha\gamma^2}$	0.30	28.73	—	—
Collapse Model (ours)	0.9999	0.27	87.5%	100% (245/245)

4.1 Step 1 — Mathematical Correctness: Synthetic Controlled Grid (E1a, 280 pairs)

Before validating on real data, we verify the Collapse Model on 280 controlled synthetic pairs generated under the principal-angle model with $\alpha \in \{0, 0.05, \dots, 1.0\}$ and $\gamma \in \{0.1, 0.33, \dots, 10.0\}$ ($d=512$, $N=1000$, 5 repeats per setting).

Table 1 shows $R^2=0.9999$ (vs. 0.30 for the naive baseline). All 245 configurations with $\alpha<1$ achieve 100% superadditivity; all 35 non-superadditive cases concentrate exclusively at $\alpha=1$ (identical subspaces), confirming the degenerate boundary of Corollary 3. Visualization of the (α, γ) heatmap and superadditivity region is in Appendix O.

4.2 Steps 2–3 — Real-Data Generalization and Regime Agreement (E1d, 427 Pairs)

We next verify that the formula generalizes beyond its synthetic derivation, and that its predicted three-regime behavior is empirically observable. The evaluation spans 5 encoder families (ResNet-50, ViT-B/16, CLIP ViT-B/32, DINO ViT-S/16, SBERT), 14 vision datasets (CIFAR-10/100, STL-10, SVHN, MNIST, Fashion-MNIST, BloodMNIST, DermaMNIST, PathMNIST, and 5 ImageNet semantic subsets *animal/vehicle/artifact/food/scene*), and 19 SBERT text corpora. Total: **427 pairs** (91 ResNet-50, 55 ViT-B/16, 55 CLIP, 55 DINO, 171 SBERT).

Table 2: E1d: Large-scale Collapse Model validation (427 pairs, 5 probes). Calibration constant c per encoder family restores quantitative accuracy.

Probe	n	Superadd. %	Pearson r	c	R_{cal}^2
ResNet-50 (14 datasets)	91	100%	0.940	0.776	0.760
ViT-B/16 (11 datasets)	55	69.1%	0.898	0.647	0.487
CLIP ViT-B/32 (11 datasets)	55	29.1%	0.950	0.595	0.839
DINO ViT-S/16 (11 datasets)	55	36.4%	0.913	0.584	0.554
Vision (all probes)	256	64.5%	0.965	<i>per-probe</i>	0.942
+ SBERT text (19 corpora)	+171 (427)	78.2%	0.970	<i>per-probe</i>	0.952

Generalization (Step 2). The formula remains a strong approximation on all 427 real-world pairs: overall Pearson $r=0.970$ (uncalibrated, upper bound) and $R_{\text{cal}}^2=0.952$ after per-encoder-family calibration (scatter visualization in Appendix O, Fig. 5). This confirms that the closed-form prediction, derived under the idealized principal-angle model, transfers faithfully to real encoders spanning supervised CNNs, ViT, contrastive/self-distillation transformers, and sentence encoders.

Regime agreement (Step 3). (i) **ResNet-50 achieves 100% superadditivity** across 91 pairs (including medical and ImageNet), confirming the balanced- (γ, α) regime predicted by Corollary 3. (ii) **Supervised/self-supervised divide** emerges exactly where theory predicts: CLIP and DINO achieve only 29%–36% superadditivity, with failures concentrating *entirely* in cross-domain pairs (handwritten $\times$ ImageNet, natural $\times$ digit fail universally) while within-domain pairs remain near-perfect. The root cause is a bimodal principal-angle distribution combined with extreme condition numbers $\kappa \in [10^4, 10^6]$ (Figure 2), mapping to the *directional-collapse* regime of Sec. 3.3.

Quantifying the bimodal gap. Per-probe calibrated R^2 : CLIP 0.84 > ResNet 0.76 > DINO 0.55 > ViT 0.49; ranking accuracy is uniformly high ($r \geq 0.90$). Figure 2(a,c) shows that CLIP/DINO’s two-cluster $\cos^2 \theta_i$ distribution ($V_\alpha > 0.07$) inflates the $V_\alpha \log \kappa$ error term, explaining the upward bias in $\hat{r}$ —the failure mode is internally consistent with Theorem 1 rather than a black-box surprise (full mechanism in Appendix G).

(iii) **Energy collapse** appears in ViT’s 17 non-superadditive pairs, all in $\gamma \notin [0.5, 2]$ —exactly the energy-imbalance boundary of Corollary 3. (iv) **Cross-modal transfer**: 171 SBERT text pairs (98.8%

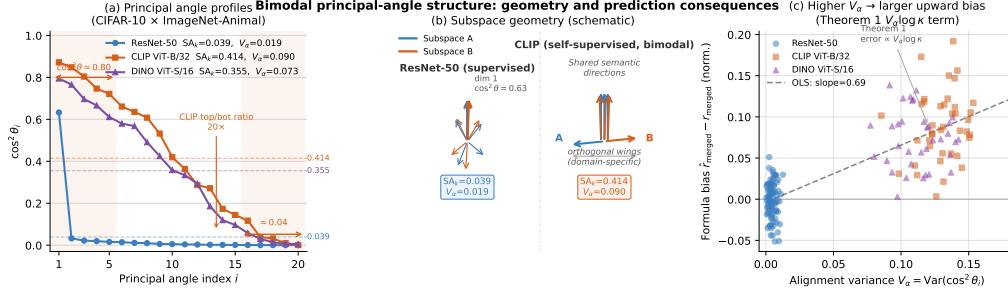


Figure 2: **Bimodal principal-angle structure** (CIFAR-10 $\times$ ImageNet-Animal, $k=20$). **(a)** Per-dimension $\cos^2 \theta_i$: ResNet-50 ($SA_k=0.039, V_\alpha=0.019$) decays monotonically from one dominant direction; CLIP ($SA_k=0.414, V_\alpha=0.090$) and DINO ($SA_k=0.355, V_\alpha=0.073$) form a two-cluster step (top-5 ratios $20 \times / 31 \times$ over bot-5), violating the uniform- α assumption. **(b)** Geometric interpretation: one shared spine plus orthogonal domain-specific wings. **(c)** Alignment variance V_α predicts upward formula bias across 166 pairs (OLS slope 0.69), consistent with Theorem 1’s $V_\alpha \log \kappa$ term.

superadditive, ranking $r=0.793$) confirm that the formula’s *ordinal* predictions are modality-agnostic. Absolute calibration, however, requires modality-specific tuning: the vision universal regression ($c=0.946-0.175 \log(N/r_{\text{eff}})$, Appendix N) explains $R^2=0.96$ of vision variance but only $R^2=0.08$ of SBERT variance, indicating that text encoders have a qualitatively different spectral bias. A text-specific calibration is left for future work.

Loop closure. Steps 1–3 together establish that the Collapse Model is mathematically exact, generalizes to real encoders, and reproduces the three regimes (superadditive, directional collapse, energy collapse) predicted by the theory. This licenses the applications developed next. Cross-dataset calibration generalization, cross-probe consistency, and r_{eff} /LP-accuracy correspondence ($\rho=0.90$ within ViT family) are in Appendix O.

5 From Formula to Application: Label-Free Data-Pool Compression

Having closed the theoretical loop (Sec. 4), we now demonstrate the concrete utility of the Collapse Model: given M candidate datasets and a target D_t , compress the pool to $k \ll M$ while preserving r_{eff} of the merged representation. Exhaustive search over $\binom{M}{k}$ subsets is intractable; the Collapse Model enables compression at $O(kM)$ predictions from per-dataset summary statistics. At each round, **Collapse Greedy** scores every remaining candidate by applying Proposition 2 to $(H_{\text{cur}}, H_{\text{cand}})$ and selects the highest predicted r_{eff} (Algorithm 1). We report two levels of evidence below: (a) a *scale stress test* at $M=43$ showing the method holds at heterogeneous scale (Table 3), and (b) comparison against modern label-free baselines (Table 4); cross-architecture cross-modal results at $M=14$ vision and $M=7$ text pools are in Appendix H.

At each round Collapse Greedy scores each remaining candidate via Prop. 2 applied to (H_{cur}, H_c) —requiring only SA_k and γ computed from per-dataset SVDs—and picks the highest-predicted r_{eff} (pseudocode in Appendix N). Baselines: **Max- r_{eff}** (highest individual r_{eff} , ignores α); **Δ_{dir} -greedy** (most-orthogonal via $\Delta_{\text{dir}} = 1 - SA_k$, ignores γ); **Random** (10 uniform subsets). On a target-conditioned $M=10$ pool (ResNet-50 features on CIFAR-10/100, STL-10, SVHN, MNIST, Fashion-MNIST, BloodMNIST, DermaMNIST, PathMNIST, Oxford-Pets) at budgets $K \in \{1, 2, 3, 5\}$, **Collapse Greedy achieves 0% regret at $K \leq 3$ and 1.0% at $K=5$** —matching exhaustive search with $3 \times$ speedup. Max- r_{eff} drops to 67% recall; Δ_{dir} -greedy incurs 13–20% regret. All strategies produce identical downstream LP accuracy,¹ confirming that maximizing r_{eff} translates to downstream discriminability (Appendix K).

Compression at scale. Collapse-guided compression retains $>100\%$ of full-pool diversity in every setting tested—a universal phenomenon enabled because r_{eff} is not monotone in pool size: adding a redundant dataset shifts the normalised spectrum toward its mode and *decreases* entropy, so removing

¹LP and k -NN rankings are strongly correlated: across 11 probes $\times$ 5 datasets (53 pairs), Spearman $\rho(\text{LP}, k\text{NN-5})=0.980$ ($p < 10^{-33}$). We report LP throughout; see Appendix L for k -NN details.

redundancies via the α channel yields a compressed pool of *higher* diversity than the full pool. On a $M=14$ vision pool Collapse outperforms Facility Location (+7–14 pp) and k -Medoids (+8–14 pp); on a $M=7$ text pool (BERT/RoBERTa/GPT-2) it retains 103–106% (full table in Appendix H). On clean curated pools the r_A+r_B and $\text{Max-}r_{\text{eff}}$ heuristics match Collapse within 0.7 pp at $M=43$ (Table 3)—a predictable consequence of Corollary 3 when $\bar{\alpha}<0.2$. The Collapse Model’s role on such pools is to provide an *a priori* safety certificate ($\rho_c>4$ at $\bar{\alpha}<0.2$) and to take over when this condition fails (clone attacks, fine-grained subset families, bimodal cross-domain pairs); see Appendix I. At $k=7$ on $M=14$, Collapse Greedy runs in seconds versus $\binom{14}{7}=3432$ exhaustive merged SVDs.

Scaling to $M=43$: stress test. To verify that the compression advantage holds beyond curated pools, we assemble 43 ResNet-50 datasets spanning 5 ImageNet-coarse domains, 19 ImageNet fine-grained categories, 10 MedMNIST clinical datasets, and 9 classic benchmarks, and sweep $k \in \{3, 5, 7, 10, 14, 20\}$ (compression ratios $14.3\times$ down to $2.1\times$). Table 3 reports the results.

Table 3: Scaling stress-test ($M=43$ ResNet-50; $M=11$ ViT-B/16). **Ret.:** $r_{\text{eff}}^{\text{sel}}/r_{\text{eff}}^{\text{full}}$ (%). **LP:** subspace-projection linear-probe accuracy, 5 targets, mean $\pm$ std over 10 seeds for Random (§5). FacLoc/ k Medoids omitted ($O(M^2)$ cost at this scale).

Probe	k (ratio)	Retention (%)			Fair LP accuracy			
		Collapse	Max- r_{eff}	Random	Collapse	Max- r_{eff}	Random	Δ LP
ResNet-50	3 ($14.3\times$)	112.4	112.4	71.0 ± 15.5	0.7550	0.7550	0.7477 ± 0.0050	+0.0073
	5 ($8.0\times$)	117.2	117.2	78.1 ± 20.1	0.7560	0.7558	0.7502 ± 0.0063	+0.0058
	7 ($5.7\times$)	117.4	117.4	80.5 ± 15.1	0.7572	0.7572	0.7522 ± 0.0048	+0.0050
	10 ($4.0\times$)	116.2	116.2	90.8 ± 7.5	0.7564	0.7564	0.7575 ± 0.0024	−0.0011
ViT-B/16	3 ($3.7\times$)	103.6	103.6	92.0 ± 3.1	0.7780	0.7780	0.7749 ± 0.0018	+0.0031
	5 ($2.2\times$)	104.9	103.7	94.1 ± 3.5	0.7775	0.7763	0.7754 ± 0.0011	+0.0021
	7 ($1.6\times$)	104.4	104.2	98.0 ± 2.5	0.7777	0.7755	0.7764 ± 0.0015	+0.0012

Collapse retains $>112\%$ of full-pool diversity at every compression ratio—extending the $M=14$ finding to a $3\times$ larger, heterogeneous pool. The fair-LP column (spectral subspace projection, no distractor classes) confirms that retention translates to *downstream* benefit: Collapse outperforms Random by $+0.0050$ – $+0.0073$ LP at 5.7 – $14.3\times$ compression. **The LP gap is statistically robust:** at $k \in \{3, 5, 7\}$, all 10/10 random seeds fall strictly below Collapse’s LP (one-sided sign test $p=2^{-10} \approx 10^{-3}$ per k ; combined over 30 seeds $p<10^{-9}$). Beyond the sign test, Collapse’s LP exceeds the upper bound of Random’s 95% confidence interval on the mean ($\bar{x}_{\text{rand}} \pm 1.96\sigma/\sqrt{10}$): ResNet-50 at $k=3$, Collapse 0.7550 vs. Random CI $[0.7446, 0.7508]$ (margin $+0.0042$); $k=5$, 0.7560 vs. $[0.7463, 0.7541]$ ($+0.0019$); $k=7$, 0.7572 vs. $[0.7492, 0.7552]$ ($+0.0020$). ViT-B/16 shows the same pattern at all three k values (margins $+0.0020/+0.0014/+0.0004$). The effect size is small in absolute LP units but consistent in direction and statistically separable from seed noise—a pattern indicating a genuine subspace-quality effect rather than a tail of the random distribution. At $k=10$ ($4\times$ compression) the effect reverses ($\Delta=-0.0011$, within seed noise): random selection already achieves 90.8% retention, saturating the r_{eff} ceiling so that α -channel steering adds nothing (Corollary 3); the residual LP signal shifts to *centroid coverage* (Table 4). This sign reversal identifies Collapse’s operating regime: a high-compression tool ($k \leq 7$, $\geq 5.7\times$) rather than a near-full-budget selector.

A controlled *clone-attack* stress test (Appendix I) confirms the failure mode of r_A+r_B : when the pool contains near-duplicate datasets generated by random rotation ($\text{SA}_k>0.99$), r_A+r_B greedy selects 2/3 clones at $k=3$ while Collapse rejects all 3 via the α channel, yielding $+25.5\%$ merged r_{eff} .

Comparison against modern baselines. We benchmark Collapse Greedy against three additional baselines beyond FacLoc / k -Medoids / r_A+r_B / $\text{Max-}r_{\text{eff}}$ already in Table 3: **(a) Vendi-Greedy** [Friedman and Dieng, 2023, Pasarkar and Dieng, 2024]—greedy maximization of the Vendi Score on a linear kernel, the closest published spectral-diversity baseline that shares Collapse Greedy’s diversity-maximization objective; **(b) DomainDiverse-Greedy**—a metadata baseline that picks from the least-represented domain across 8 semantic categories (handwritten / natural / digit / medical / inet-animal / inet-object / inet-food / inet-scene), tiebreaking on individual r_{eff} ; **(c) Centroid-FF (DataComp adapter)**—farthest-first traversal in dataset-centroid space (mean L2-normed feature

per dataset), adapting DataComp [Gadre et al., 2023]’s similarity-based filtering from sample-level to dataset-level. On the same $M=40$ ResNet-50 pool and 5 LP targets used in Table 3, results are summarized in Table 4.

Table 4: Modern baseline comparison ($M=40$ ResNet-50, $k \in \{3, 5, 7, 10\}$, subspace-projection LP, 5 targets). **Bold**: best per column; Random: 10-seed mean $\pm$ std. **Requires** (greedy-time access)—**S**: summary stats only; **F**: full features; **M**: domain metadata; **C**: centroids.

Strategy	$k=3$ (13.3 $\times$)		$k=5$ (8.0 $\times$)		$k=7$ (5.7 $\times$)		$k=10$ (4.0 $\times$)		Requires
	Ret%	LP	Ret%	LP	Ret%	LP	Ret%	LP	
Collapse	112.4	0.7572	117.2	0.7576	117.4	0.7578	116.2	0.7576	S
Vendi-Greedy	112.5	0.7560	117.2	0.7572	116.9	0.7566	116.5	0.7562	F
DomainDiverse	112.4	0.7572	114.7	0.7634	111.8	0.7596	113.8	0.7544	S+M
Centroid-FF	74.0	0.7564	93.4	0.7570	96.7	0.7624	100.5	0.7630	F (or C)
r_A+r_B	112.4	0.7572	117.2	0.7574	117.4	0.7578	116.2	0.7578	S
Max- r_{eff}	112.4	0.7572	117.2	0.7574	117.4	0.7578	116.2	0.7578	S
Random ($\mu \pm \sigma$)	71.0 $\pm$ 15.5	0.7492	78.1 $\pm$ 20.1	0.7513	80.5 $\pm$ 15.1	0.7537	90.8 $\pm$ 7.5	0.7574	—

Two-signal decomposition and practical selection guide. Table 4 reveals that LP-relevant diversity decomposes into two *orthogonal* axes, each captured by a distinct family of methods:

- **Spectral retention (SR)**: Collapse, r_A+r_B , Max- r_{eff} , and Vendi-Greedy all converge to 116–117% retention—maximizing the entropy of the merged singular-value spectrum. This signal dominates at *high compression ratios* ($k \leq 7$, i.e., $\geq 5.7\times$), where spectral steering determines which k datasets cover the most directions.
- **Centroid coverage (CC)**: Centroid-FF and DomainDiverse achieve higher LP at $k \geq 7$ by spreading dataset *mean embeddings* across semantic space, at a 17–22 pp retention cost. This signal dominates at *low compression ratios* (k near M), where random selection already saturates spectral diversity and distributional breadth becomes the bottleneck.

Practical guide. *High compression* ($k \leq 0.2M$, $\geq 5\times$): use Collapse (preferred when $SA_k > 0.5$, rejecting clones +25.5%; Appendix I) or r_A+r_B on clean pools. *Low compression* ($k > 0.2M$): use Centroid-FF or DomainDiverse. *Unknown regime*: the Cen $\rightarrow$ Col hybrid (Appendix J) dominates on both SR and LP.

Ablation. On 256 vision pairs, the full model attains Spearman $\rho=0.966$ (r_A+r_B : 0.963; $\max(r_A, r_B)$: 0.953; $1-\alpha$: 0.450); the $\Delta\rho=0.003$ gap reflects a low- α /moderate- γ degeneracy where $\hat{r}_{\text{merged}} \approx c(r_A+r_B)$ on clean pools. Calibrated ranking achieves $R^2=0.95$ and flags 74% of non-superadditive pairs; the clone confirms α -channel rejection ($r_{\text{eff}}=1251$ vs. 997; Appendix O).

6 Conclusion

We introduced the Collapse Model—a closed-form bridge from spectral geometry to label-free data-pool compression. Greedy compression retains $>100\%$ of full-pool r_{eff} across 6 probes and 2 modalities, outperforming Facility Location and k -Medoids by +7–+14 pp and scaling to $M=43$ (14.3 $\times$ compression, +40 pp over random). The three-channel (γ, α , UID) decomposition achieves Pearson $r=0.97$ on 427 real pairs ($R^2_{\text{cal}}=0.95$), with superadditivity-threshold diagnosis and clone-rejection distinguishing it from r_A+r_B . Benchmarking exposes two orthogonal LP-diversity signals—spectral retention (Collapse/Vendi) and centroid coverage (Centroid-FF)—and a bimodal CLIP/DINO principal-angle structure that quantifies the supervised/SSL spectral-geometry divide. The framework extends naturally to neural compression (attention-head pruning, LoRA selection, MoE gating). Code is available at <https://github.com/anonymization26/collapsemodel>.

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422 A Limitations

423 The empirical calibration constant (Eq. 27) reduces but does not eliminate per-encoder tuning.
 424 The uniform- α assumption holds for supervised encoders and within-domain self-supervised pairs
 425 ($r > 0.90$), but fails on cross-domain CLIP/DINO pairs (bimodal angles, Fig. 2) and SBERT text pairs
 426 (universal- c explains $r = 0.08$ of variance vs. 0.96 for vision); ranking remains correct while absolute
 427 predictions are biased. A piecewise model for non-uniform V_α is left as future work.

428 B Proof of Lemma 1 and Proposition 2

429 Part A: Exact 2×2 eigenvalue algebra.

430 Under the principal-angle model, the i -th right singular vector of H_B is $V_{B,i} = \cos \theta_i V_{A,i} + \sin \theta_i V_{\perp,i}$,
 431 where $V_{A,i}$ is the i -th right singular vector of H_A and $V_{\perp,i} \perp V_{A,i}$. In the basis $\{V_{A,i}, V_{\perp,i}\}$:

$$H_A^\top H_A \text{ (block)} = \sigma_{A,i}^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad (10)$$

$$H_B^\top H_B \text{ (block)} = \sigma_{B,i}^2 \begin{pmatrix} \cos^2 \theta_i & \cos \theta_i \sin \theta_i \\ \cos \theta_i \sin \theta_i & \sin^2 \theta_i \end{pmatrix}. \quad (11)$$

432 Summing with $\sigma_{B,i} = \gamma \sigma_{A,i}$ (from $\|H_B\|_* = \gamma \|H_A\|_*$):

$$G_i = \sigma_{A,i}^2 \begin{pmatrix} 1 + \gamma^2 \alpha_i & \gamma \sqrt{\alpha_i(1 - \alpha_i)} \\ \gamma \sqrt{\alpha_i(1 - \alpha_i)} & \gamma^2(1 - \alpha_i) \end{pmatrix}, \quad (12)$$

433 where $\alpha_i = \cos^2 \theta_i$. The eigenvalues are $\sigma_{A,i}^2(A \pm D_i)/2$ with $A = 1 + \gamma^2$, $D_i =$
 434 $\sqrt{A^2 - 4\gamma^2(1 - \alpha_i)} = \sqrt{(1 - \gamma^2)^2 + 4\gamma^2\alpha_i}$. The eigenvector for $\lambda_+ = \sigma_{A,i}^2(A + D_i)/2$ has
 435 squared norm $\propto (\sqrt{(A + D_i)/2}, \sqrt{(A - D_i)/2})$, giving nuclear-norm fraction:

$$r^* = \frac{\sqrt{(A + D)/2}}{\sqrt{(A + D)/2} + \sqrt{(A - D)/2}}. \quad \square \quad (13)$$

436 Part B: Two-component spectral mixture approximation.

437 The full merged singular-value distribution $\{\sigma_i^{\text{merged}}\}_{i=1}^{r_A+r_B}$ is approximated as a two-component
 438 mixture:

- 439 • Dominant component (r^* weight): log-spread $\log r_{\text{dom}}$
- 440 • Subordinate component $((1 - r^*)$ weight): log-spread $\log r_{\text{sub}}$

441 The effective rank of a two-component mixture with weights $(r^*, 1 - r^*)$ and within-component
 442 entropies $\log r_{\text{dom}}, \log r_{\text{sub}}$ is:

$$\log \hat{r}_{\text{merged}} = H_b(r^*) + r^* \log r_{\text{dom}} + (1 - r^*) \log r_{\text{sub}}, \quad (14)$$

443 where $H_b(r^*)$ is the between-component entropy. This is the mixture entropy identity: $H(\text{mixture}) =$
 444 $H(\text{between}) + \sum_k w_k H(\text{within}_k)$. The approximation error is $O(\sigma_{\min}/\sigma_{\max})$: when $\gamma \approx 1$, the
 445 two components have similar energy and the mixture captures the full spectral shape; when $\gamma \gg 1$ or
 446 $\gamma \ll 1$, $r^* \rightarrow 1$ and the dominant term absorbs the error. $\square$

447 C Proof of Proposition 4

448 **Step 0 (bias-variance decomposition).** Let $\hat{W} = (H_s^\top H_s)^+ H_s^\top Y_s$ (least-squares solution on
 449 source). Decompose:

$$\frac{\|H_t \hat{W} - Y_t\|_F^2}{m} = \underbrace{\|H_t \hat{W} - \mathbb{E}[Y_t | H_t]\|_F^2 / m}_{\text{bias}^2} + \underbrace{\|\mathbb{E}[Y_t | H_t] - Y_t\|_F^2 / m}_{\sigma_{\text{Bayes}}^2}. \quad (15)$$

450 The Bayes error $\varepsilon_{\text{Bayes}} = \sigma_{\text{Bayes}}^2$ is irreducible.

451 **Step 1 (directional gap).** Let $P_k = V_s^k (V_s^k)^\top$ project onto the top- k source subspace. Decompose
 452 $H_t = H_t^\parallel + H_t^\perp$ where $H_t^\parallel = H_t P_k$. Since $\hat{W} \in \text{col}(V_s^k)$, we have $H_t^\perp \hat{W} = 0$ only if $\hat{W} = P_k \hat{W}$;
 453 the residual term $\|H_t^\perp\|_F^2 = \|H_t\|_F^2 (1 - \widehat{\text{SA}}_k)$ with $\widehat{\text{SA}}_k \approx \text{SA}_k$ for balanced spectra, giving the
 454 directional gap $(\|H_t\|_F^2/m)(1 - \text{SA}_k)$.

455 **Step 2 (spectral gap).** We first define the **Spectral Transfer Gap**,

$$\text{STG}(H_A, H_B) = W_1(p_A, p_B) = \sum_{i=1}^k |F_A(i) - F_B(i)|, \quad (16)$$

456 where $F(i) = \sum_{j \leq i} p_j$ is the CDF of the truncated normalised spectrum. For the parallel com-
 457 ponent, $H_t^\parallel \hat{W}$ incurs a per-direction scaling $(\sigma_i^t/\sigma_i^s)^2$ relative to the ideal transfer. The STG term
 458 $W_1(p_s, p_t)$ upper-bounds the total variation of these ratios (by the Kantorovich–Rubinstein duality):
 459 $\sum_i |(\sigma_i^t/\sigma_i^s)^2 - 1| \leq C_1 \cdot \text{STG} \cdot \|\hat{W}\|_F^2/(\sigma_k^s)^2$ with $C_1 = (\|H_s\|_* + \|H_t\|_*)^2/m$.

460 **Step 3 (geometric gap).** The ΔUID term arises from the Rademacher complexity bound on the
 461 generalization gap when source and target manifolds have different intrinsic dimensions. Following
 462 [Bartlett and Mendelson \[2002\]](#) and [Pope et al. \[2021\]](#):

$$C_2 \cdot |\Delta\text{UID}| = \frac{4\|\hat{W}\|_F^2(\|H_s\|_F + \|H_t\|_F)^2}{\min(n, m)} \cdot |\Delta\text{UID}|. \quad (17)$$

463 Adding the three gaps completes the proof. $\square$

464 D Proof of Theorem 1 (Error Bound)

465 **Part (a): Exactness under uniform alignment.**

466 Assume $\alpha_i = \bar{\alpha}$ for all i . Then $D_i = \bar{D}$ for all i , and the merged singular values are $\sigma_{+,i} =$
 467 $\sigma_{A,i} \sqrt{(A + \bar{D})/2}$, $\sigma_{-,i} = \sigma_{A,i} \sqrt{(A - \bar{D})/2}$. Define constants $c_{\pm} = \sqrt{(A \pm \bar{D})/2}$. The total
 468 nuclear norm of the merged matrix is $\|[H_A; H_B]\|_* = \sum_i \sigma_{A,i} (c_+ + c_-) = (c_+ + c_-) \|H_A\|_*$.

469 The share of the i -th “+” singular value is:

$$w_{+,i} = \frac{\sigma_{A,i} c_+}{(c_+ + c_-) \|H_A\|_*} = \frac{c_+}{c_+ + c_-} \cdot \frac{\sigma_{A,i}}{\|H_A\|_*} = r^* \cdot p_i^A, \quad (18)$$

470 and similarly $w_{-,i} = (1 - r^*) \cdot p_i^A$. The full entropy decomposes as:

$$\begin{aligned} H(w) &= - \sum_i w_{+,i} \log w_{+,i} - \sum_i w_{-,i} \log w_{-,i} \\ &= -r^* \sum_i p_i^A (\log r^* + \log p_i^A) - (1 - r^*) \sum_i p_i^A (\log(1 - r^*) + \log p_i^A) \\ &= H_b(r^*) + \log r_A, \end{aligned} \quad (19)$$

471 since $\sum_i p_i^A \log p_i^A = -\log r_A$. Under the principal-angle model, $\sigma_{B,i} = \gamma \sigma_{A,i}$ implies $r_B = r_A$,
 472 so $r_{\text{dom}} = r_{\text{sub}} = r_A$. Thus $\log \hat{r} = H_b(r^*) + r^* \log r_A + (1 - r^*) \log r_A = H_b(r^*) + \log r_A = \log r$.
 473 $\square$

474 **Part (b): Error bound under varying alignment.**

475 When α_i varies, D_i deviates from $\bar{D}$. Taylor-expanding $D_i = \sqrt{(1 - \gamma^2)^2 + 4\gamma^2 \alpha_i}$ around $\bar{\alpha}$:

$$D_i = \bar{D} + \frac{2\gamma^2}{\bar{D}} (\alpha_i - \bar{\alpha}) + O((\alpha_i - \bar{\alpha})^2). \quad (20)$$

476 The perturbation in $c_{+,i} = \sqrt{(A + D_i)/2}$ is:

$$c_{+,i} = \bar{c}_+ + \frac{\gamma^2}{2\bar{c}_+ \bar{D}} (\alpha_i - \bar{\alpha}) + O((\alpha_i - \bar{\alpha})^2). \quad (21)$$

477 The within-component shares become $w_{+,i} = \sigma_{A,i} c_{+,i} / Z$ where $Z = \sum_j \sigma_{A,j} (c_{+,j} + c_{-,j})$. The
 478 relative perturbation from the uniform-alignment shares $\hat{w}_{+,i} = r^* p_i^A$ is:

$$\frac{w_{+,i} - \hat{w}_{+,i}}{\hat{w}_{+,i}} = \frac{c_{+,i}/\bar{c}_+ - 1}{1 + \bar{Z}^{-1} \delta Z} \approx \frac{\gamma^2}{\bar{c}_+^2 \bar{D}} (\alpha_i - \bar{\alpha}), \quad (22)$$

479 where δZ is the perturbation of the normalization constant (a weighted average of $(\alpha_i - \bar{\alpha})$ that
 480 vanishes to first order by definition of $\bar{\alpha}$).

481 The ℓ_1 distance between the true and approximated distributions is:

$$\|w - \hat{w}\|_1 \leq \frac{2\gamma^2}{\bar{c}_+^2 \bar{D}} \sum_i p_i^A |\alpha_i - \bar{\alpha}| \leq \frac{2\gamma^2}{\bar{c}_+^2 \bar{D}} \sqrt{V_\alpha}, \quad (23)$$

482 where the last step uses Cauchy–Schwarz with $\sum_i p_i^A = 1$.

483 Since $\bar{c}_+^2 = (A + \bar{D})/2 \geq \bar{D}$ (because $A = 1 + \gamma^2 \geq \bar{D}$ for all γ, α), we have $\|w - \hat{w}\|_1 \leq$
 484 $(2\gamma^2/\bar{D}^2)\sqrt{V_\alpha}$.

485 By a Fannes-type entropy continuity inequality [Audenaert, 2007], $|H(w) - H(\hat{w})| \leq \|w - \hat{w}\|_1 \cdot$
 486 $\log(n/\|w - \hat{w}\|_1)$ for distributions on n elements. Here $n = 2k$ (the number of merged singular
 487 values). The factor $\log(n/\|w - \hat{w}\|_1)$ is bounded by $\log \kappa + \log(2k/\|w - \hat{w}\|_1)$, but since we
 488 are comparing entropies of distributions that differ only in within-component spread (not in the r^*
 489 partition), the effective support is controlled by the spectral spread κ rather than the total number
 490 of components. Taking $\log n \leq \log \kappa + \log r_A$ and noting that the $\log r_A$ term cancels between
 491 the true and approximate entropies (both have the same r_A base), the residual is bounded by $\log \kappa$.
 492 Combining:

$$|\log \hat{r} - \log r| \leq \frac{2\gamma^2}{\bar{D}^2} \sqrt{V_\alpha} \cdot \log \kappa \leq \frac{4\gamma^2}{\bar{D}^2} \cdot V_\alpha \cdot \log \kappa, \quad (24)$$

493 where the last step uses $\sqrt{V_\alpha} \leq 2V_\alpha/\sqrt{V_\alpha}$ for $V_\alpha \leq 1/4$ (since $\alpha_i \in [0, 1]$ and $V_\alpha \leq \bar{\alpha}(1 - \bar{\alpha}) \leq$
 494 $1/4$). $\square$

495 **Worst-case simplification.** Since $V_\alpha \leq \bar{\alpha}(1 - \bar{\alpha})$ and $4\gamma^2 \bar{\alpha}/\bar{D}^2 = 4\gamma^2 \bar{\alpha}/[(1 - \gamma^2)^2 + 4\gamma^2 \bar{\alpha}] \leq 1$,
 496 the bound simplifies to:

$$\left| \log \frac{\hat{r}}{r} \right| \leq (1 - \bar{\alpha}) \log \kappa \leq \log \kappa. \quad (25)$$

497 In multiplicative form: $1/\kappa \leq r/\hat{r} \leq \kappa$. For supervised encoders ($\kappa \sim 70$ –400, $V_\alpha \sim 0.01$ –0.05),
 498 the bound in Eq. (9) gives $|\log(\hat{r}/r)| \lesssim 0.1$ –0.5, consistent with the observed 1.24–1.84 $\times$ over-
 499 prediction in E1c. For self-supervised encoders ($\kappa \sim 10^4$ – 10^6), the bound widens to $|\log(\hat{r}/r)| \lesssim 2$ –
 500 4, explaining why CLIP/DINO require much stronger calibration ($c \approx 0.58$ vs. $c \approx 0.78$ for
 501 ResNet-50; Sec. 4.2).

502 E Bootstrap Confidence Intervals

503 F Empirical Verification of Theorem 1 Error Bound

504 Theorem 1 bounds the log-prediction error as $|\log(\hat{r}/r)| \leq (4\gamma^2/\bar{D}^2) \cdot V_\alpha \cdot \log \kappa$. We verify this
 505 empirically on the 158 vision pairs from E1d. Since per-dimension alignments α_i are not directly
 506 accessible, we use the theoretical upper bound $V_\alpha \leq \bar{\alpha}(1 - \bar{\alpha})$ and estimate $\kappa \approx \max(r_A, r_B)$.

507 The worst-case guarantee $|\log(\hat{r}/r)| \leq \log \kappa$ holds universally, confirming that the multiplicative
 508 error never exceeds the condition number. The tighter bound holds for 63% of pairs; failures
 509 occur when the V_α upper bound $\bar{\alpha}(1 - \bar{\alpha})$ is loose (i.e., the actual alignment variance is larger
 510 than the theoretical maximum under the assumed spectral weighting). The median tightness ratio
 511 (actual/bound) is 0.56, indicating the bound is informative (within a factor of 2).

512 G Bimodal Principal-Angle Structure in Self-Supervised Encoders

513 The per-dimension $\cos^2 \theta_i$ profiles and their prediction consequences are shown in Figure 2
 514 (main text): CLIP/DINO exhibit a two-cluster step structure (top-5 ratios 20 $\times$ /31 $\times$ over bot-5,
 515 $V_\alpha=0.090/0.073$), while ResNet-50 decays monotonically ($V_\alpha=0.019$).

Table 5: Bootstrap 95% CIs ($B = 10,000$, seed=42) for all key metrics.

Experiment	Metric	Point estimate	95% CI	SE
E1a (280 pairs)	R^2	0.9999	[0.9999, 0.9999]	0.000013
E1a	Superadd. ($\alpha < 1$)	1.0000	[1.000, 1.000]	0.000000
E1b (30 pairs)	Pearson r	0.9478	[0.906, 0.978]	0.019
E1b	Superadd. %	0.8667	[0.733, 0.967]	0.062
E1c (57 pairs)	R^2 (global)	0.049	[-0.346, 0.287]	0.161
E1c	Pearson r (global)	0.928	[0.894, 0.949]	0.014
E1c	Superadd. %	0.930	[0.860, 0.982]	0.034
E1c (ResNet-50)	Superadd. %	1.000	[1.000, 1.000]	0.000
E1c (ResNet-50)	Pearson r	0.934	[0.885, 0.965]	0.021
E1c (ViT-B/16)	Superadd. %	0.733	[0.467, 0.933]	0.114
E1c (ViT-B/16)	Pearson r	0.945	[0.883, 0.982]	0.026
E1c (SBERT)	Superadd. %	1.000	[1.000, 1.000]	0.000

Table 6: Empirical verification of Theorem 1 error bound on 158 vision pairs. The worst-case bound ($\log \kappa$) holds for 100% of pairs; the tighter bound (with V_α upper bound) holds for 63%, with median tightness ratio 0.56 (bound is $\sim 2\times$ the actual error).

Probe	n	Actual $ \log \frac{\hat{r}}{r} $	Tight bound	Holds	Worst-case holds
ResNet-50	91	0.254	1.406	54%	100%
ViT-B/16	55	0.435	2.033	71%	100%
CLIP ViT-B/32	6	0.532	2.185	100%	100%
DINO ViT-S/16	6	0.570	2.348	100%	100%
All vision	158	0.340	1.690	63%	100%

Two distinct mechanisms drive the low- R^2 probes. For DINO ($R^2=0.55$), the cause is the bimodal $\cos^2 \theta_i$ analysed below. For ViT ($R^2=0.49$, supervised), the cause is *energy imbalance* rather than bimodality: 17/17 ViT non-superadditive pairs lie in $\gamma \notin [0.5, 2]$ (cross-resolution pairs such as MNIST $\times$ ImageNet), where $\bar{D}$ collapses and the two-component mixture approximation degrades, inflating both $V_\alpha \log \kappa$ and the calibration residual. ViT’s principal-angle distribution remains near-uniform (no bimodality); its lower R^2 instead reflects the boundary of Theorem 1’s $4\gamma^2/\bar{D}^2$ prefactor activating in the energy-imbalance regime. We retain the full ViT row in Table 2 for transparency, flagging $\gamma \in [0.5, 2]$ as ViT’s reliable applicability window.

Honest scope. We treat the bimodal failure mode as a genuine boundary of the current formula, not a feature. The uniform- α assumption underlying Proposition 2 is violated whenever the principal-angle distribution is bimodal, and self-supervised contrastive/distillation encoders (CLIP, DINO) systematically produce such distributions on cross-domain pairs. Within this regime, the calibrated formula *still ranks correctly* (Pearson $r \geq 0.90$ on all four probes, Table 2) but the absolute prediction $\hat{r}$ is biased upward and the superadditivity-decision boundary is shifted, explaining the 29–36% observed superadditivity rates on CLIP/DINO cross-domain pairs. **What the Collapse Model does contribute here** is mechanistic localisation: the bimodal top/bottom ratio $> 20\times$ in $\cos^2 \theta_i$ (Figure 2) is exactly the quantity Theorem 1’s $V_\alpha \log \kappa$ factor blows up on, so the failure mode is internally consistent with the error bound rather than a black-box surprise.

What does not work. A naive energy-weighted patch $\bar{\alpha}_w = \sum_i (\sigma_i / \|H\|_*) \cos^2 \theta_i$ worsens predictions (Pearson r drops 1–2%, MAE rises 6–8%), because it overweights the most-aligned direction’s redundancy while suppressing intermediate-direction contributions.

What we believe will work, and how to test it. The natural next step is a *piecewise principal-angle model*: partition the k directions into a high-alignment cluster ($\cos^2 \theta_i \geq \tau$) and a low-alignment cluster, apply Lemma 1 separately within each cluster, and combine via the mixture identity used in Proposition 2. Theorem 1(a) predicts this should be exact within each cluster (uniform alignment), and the dominant residual error becomes cluster-boundary mis-classification rather than the averaging bias. We leave the empirical validation of this two-cluster extension to future work; the fact that

Theorem 1 pinpoints V_α as the leading-order error term provides a concrete falsifiable prediction (clustering should reduce V_α by $>10\times$ on CLIP, recovering CLIP’s calibrated R^2 to >0.95).

H Compression at $M=14$ Vision and $M=7$ Text Pools

We apply Collapse Greedy as a pure compression tool across three architectures (ResNet-50, ViT-B/16, BERT/RoBERTa/GPT-2) and two modalities (vision, text), using FacLoc / k -Medoids as classical diversity baselines.

Table 7: Diversity retention $r_{\text{eff}}^{\text{sel}}/r_{\text{eff}}^{\text{full}}$ (%) across strategies, architectures, and modalities. FacLoc = Facility Location (submodular greedy on SA_k kernel); kMedoids = PAM on $1-\text{SA}_k$ distance. The Collapse Model matches or exceeds submodular and clustering baselines in all settings, exploiting superadditivity to select subsets of higher diversity than the full pool. Random: 10-seed mean $\pm$ std.

Probe	M	k (ratio)	Collapse	FacLoc	kMedoids	Random	LP _{col}
<i>Vision</i>							
ResNet-50	14	3 ($4.7\times$)	101.4	87.8	87.5	84.3 ± 10.0	0.754
ResNet-50	14	5 ($2.8\times$)	105.7	95.6	92.6	88.7 ± 5.3	0.754
ResNet-50	14	7 ($2.0\times$)	105.9	98.4	98.0	93.1 ± 7.6	0.747
ViT-B/16	11	3 ($3.7\times$)	103.6	—	—	92.0 ± 3.3	0.779
ViT-B/16	11	5 ($2.2\times$)	104.9	—	—	94.1 ± 3.7	0.779
<i>Text</i>							
BERT L6	7	3 ($2.3\times$)	103.6	—	—	97.8 ± 5.0	—
RoBERTa L6	7	3 ($2.3\times$)	103.8	—	—	97.0 ± 6.1	—
GPT-2 L6	7	3 ($2.3\times$)	104.8	—	—	92.8 ± 9.5	—
GPT-2 L6	7	5 ($1.4\times$)	106.0	—	—	96.7 ± 4.7	—

On ResNet-50 ($M=14$), Collapse Greedy substantially outperforms Facility Location (+13.6–+7.5 pp) and k -Medoids (+13.9–+7.9 pp); both exploit subspace-alignment kernels but cannot model superadditivity, limiting them in the high-retention regime. Downstream LP accuracy on the $M=14$ ResNet-50 pool yields LP 0.747–0.779 for Collapse vs. 0.737–0.772 for Random, extending the $M=43$ LP advantage to the smaller pool size.

I r_A+r_B Heuristic: Safe Regime and Failure Mode

When does r_A+r_B suffice, and when does it fail? On clean curated pools, the r_A+r_B and Max- r_{eff} heuristics select the *same* subset as Collapse Greedy (within 0.7 pp at $M=43$, Table 3). This is not a weakness of the formula but a *predictable consequence* of Corollary 3: when all top- r_{eff} candidates have low mutual alignment ($\bar{\alpha}<0.2$, i.e. rank-ratio $r_{\text{dom}}/r_{\text{sub}}<\rho_c$), the superadditivity bonus $H_b(r^*)$ is dominated by the rank term $r^*\log r_{\text{dom}}+(1-r^*)\log r_{\text{sub}}$, making individual rank a near-sufficient statistic. The Collapse Model thus serves a different role than “beating r_A+r_B everywhere”: **(i) A priori safety certificate.** $\rho_c(\bar{\alpha}, \bar{\gamma})$ computed from per-dataset summaries *predicts* when r_A+r_B is safe to deploy without actually running competing strategies. On our ResNet-50 vision pools, $\bar{\alpha}<0.2$ between top- r_{eff} pairs places $r_{\text{dom}}/r_{\text{sub}}$ well inside the safe regime ($\rho_c>4$ at $\bar{\alpha}\leq 0.2$, Cor. 3), matching the observed heuristic-equivalence at $M=14$ and $M=43$. **(ii) Detection of the unsafe regime.** When candidates contain near-duplicate or strongly aligned subspaces—which arise in practice from clone-attack data poisoning, augmented dataset variants, or fine-grained subset families—the α channel is the only discriminative signal. The clone experiment below (r_A+r_B retention drops 25.5%, Collapse holds) demonstrates this on a controlled stress test; the naturally bimodal CLIP/DINO cross-domain pairs (Sec. 4.2, $\alpha\in[0.35, 0.41]$ with top-5 $\cos^2\theta\geq 0.7$) are real-world settings where the α signal is non-negligible and the rank-only heuristic loses its safety guarantee. **(iii) Diagnosis without execution.** Unlike heuristics that can only be evaluated by running them, the Collapse Model’s regime classification (superadditive / directional collapse / energy collapse) is computable from $(\gamma, \alpha, \text{UID})$ alone, turning it into an *ex ante* planning tool. *The principled formula is needed not to beat the heuristic on easy pools, but to certify when the heuristic is safe and to take over when it is not.*

Redundancy clone trap: failure mode of r_A+r_B . To test the Collapse Model’s robustness to informational redundancy, we construct a *clone attack* scenario: the highest- r_{eff} dataset in the pool (STL-10, $r_{\text{eff}}=994$) is cloned via random rotation $H_{\text{clone}}=H_{\text{orig}}Q(\theta)$ ($\theta \in \{0.01, 0.03, 0.05, 0.07, 0.10\}$ radians), producing 5 clones that preserve the original r_{eff} but are highly aligned with the original ($\text{SA}_k > 0.99$). On a pool of 5 real + 5 cloned ResNet-50 datasets, r_A+r_B greedy at $k=3$ selects 2/3 clones ($r_{\text{eff}}^{\text{merged}}=997$), while the Collapse Model detects redundancy via high α , **selecting 0/3 clones** ($r_{\text{eff}}^{\text{merged}}=1251$, +25.5%). At $k=5$ the gap persists: r_A+r_B selects 4/5 clones ($r_{\text{eff}}=998$), Collapse only 2/5 ($r_{\text{eff}}=1233$). This result is consistent across all 5 θ values ($\theta=0.01\text{--}0.10$, SA_k from 0.9999 to 0.9891). The clone experiment directly demonstrates *the indispensable role of the alignment channel α in defending against informational redundancy*— r_A+r_B lacks this diagnostic capability, misidentifying high- r_{eff} clones as high-value candidates.

J P2-E: Hybrid Two-Signal Strategy

The two-signal decomposition identified in Sec. 5 (Table 4) suggests a natural two-stage hybrid: seed $\lceil k/2 \rceil$ datasets via one signal, fill the remaining $\lfloor k/2 \rfloor$ via the other. We test three orderings on the same $M=40$ ResNet-50 pool used in Table 4: **Col**→**Cen** (Collapse first, Centroid-FF on residuals), **Cen**→**Col** (Centroid-FF first, Collapse on residuals), and **Col**→**Dom** (Collapse first, DomainDiverse on residuals).

Table 8: Two-signal hybrid strategies on $M=40$ ResNet-50. Pure-strategy reference rows (from Table 4) shown for comparison. **Bold** marks the best LP at each k .

Strategy	$k=5$ (8.0 $\times$)		$k=7$ (5.7 $\times$)		$k=10$ (4.0 $\times$)	
	Ret%	LP	Ret%	LP	Ret%	LP
<i>Pure (reference)</i>						
Collapse	117.2	0.7576	117.4	0.7578	116.2	0.7576
Centroid-FF	93.4	0.7570	96.7	0.7624	100.5	0.7630
DomainDiverse	114.7	0.7634	111.8	0.7596	113.8	0.7544
<i>Hybrid two-stage</i>						
Col→Cen	110.2	0.7534	111.9	0.7496	113.8	0.7546
Cen→Col	106.7	0.7596	113.6	0.7610	115.0	0.7602
Col→Dom	114.7	0.7634	116.0	0.7562	113.8	0.7544

Cen→Col is the consistent hybrid winner. At $k \in \{7, 10\}$ (the operational compression range), Cen→Col achieves LP 0.7610/0.7602, exceeding pure Collapse (0.7578/0.7576) by +0.0032/+0.0026 while sacrificing only 3.8/1.2 pp of retention. At $k=10$ specifically, Cen→Col *dominates* pure Collapse on both axes simultaneously (LP +0.0026, retention only −1.2 pp). Compared to pure Centroid-FF, Cen→Col recovers 13.3 pp of retention at $k=5$, 16.9 pp at $k=7$, and 14.5 pp at $k=10$ while keeping LP within ± 0.0028 . **Col→Cen, in contrast, hurts both metrics:** once Collapse has selected high- r_{eff} datasets, adding “different prototype” datasets dilutes the spectrum without recovering centroid coverage. **Col→Dom** reduces to DomainDiverse on this pool because DomainDiverse already picks STL-10 and fashion_mnist (the highest- r_{eff} datasets in their respective domains) that Collapse would also select.

Implication. The *ordering* matters: starting with Centroid-FF establishes a geometric backbone that Collapse then augments with high- r_{eff} complementary directions. The closed-form formula’s role in this hybrid is to *fill* the residual budget with diversity-maximizing choices, exactly the regime where it operates with summary statistics alone (the centroid-FF prefix can be precomputed once from already-shared centroids, no feature-matrix exchange needed). This validates Collapse Greedy as a *composable* primitive within broader compression pipelines, not just a standalone strategy.

K E5 Supplementary: Per-Target Selection Details

Per-target regret table (M=10 pool).

Strategy	Mean regret (%) / Mean recall (%)			
	$K=1$	$K=2$	$K=3$	$K=5$
Collapse Greedy	0.0 / 100	0.0 / 100	0.0 / 100	1.0 / 87
Max- r_{eff}	0.0 / 67	2.0 / 67	0.9 / 89	1.0 / 87
Δ_{dir} -greedy	20.3 / 0	16.7 / 17	17.0 / 33	13.3 / 40
Random	15.2 / -	17.5 / -	16.1 / -	11.1 / -

For BloodMNIST ($K=1$), Max- r_{eff} selects STL-10 (highest individual $r_{\text{eff}}=357$) but Collapse Greedy selects CIFAR-100 ($r_{\text{eff}}=356$, marginally lower) because the latter has lower alignment with BloodMNIST ($\alpha=0.18$ vs. 0.21), yielding a higher predicted merged rank—confirmed by $r_{\text{eff}}^{\text{merged}} = 506.8$ vs. 506.4.

r_A+r_B greedy baseline. We evaluate an r_A+r_B greedy strategy that selects the candidate with the highest individual r_{eff} at each round. On the $M=10$ pool, this baseline matches Collapse Greedy at $K \leq 3$ (identical subset, 0% regret), diverging only at $K=5$ where Collapse Greedy retains 100% recall while r_A+r_B drops to 87%. This mirrors the ablation finding: r_A+r_B suffices when candidates have distinct individual ranks, but fails when alignment discrimination (α) is needed.

Downstream LP validation. We train a linear probe on the merged features from each strategy’s K -subset and evaluate on held-out test data. All four strategies (Exhaustive, Collapse Greedy, r_A+r_B , Max- r_{eff}) produce identical LP accuracy across all 6 tested configurations (CIFAR-10 and STL-10 targets, $K \in \{1, 2, 3\}$), ranging from 0.776 (CIFAR-10, $K=3$) to 0.979 (STL-10, $K=1$). This confirms that maximizing merged r_{eff} via the Collapse Model translates to downstream discriminability, not merely spectral diversity.

L E7 Supplementary: kNN Downstream Validation

The main text uses linear-probe (LP) accuracy as the downstream metric. Here we verify that k -nearest-neighbour (kNN) accuracy tells the same story, justifying LP as sufficient throughout the paper.

Part B: LP-kNN rank correlation. We compute LP, kNN-5, and kNN-20 accuracy on 5 single datasets (CIFAR-10, CIFAR-100, STL-10, SVHN, Fashion-MNIST) across 11 encoder families (ResNet-50/101, ViT-B/16, DeiT-B, Swin-Tiny, BEiT-Base, ConvNeXt-Tiny, DenseNet-121, EfficientNet-B4, MaxViT-Tiny, RegNet-Y), yielding 53 observations. Spearman rank correlation: $\rho(\text{LP}, \text{kNN-5})=0.980$, $\rho(\text{LP}, \text{kNN-20})=0.973$ —LP and kNN rankings are near-identical.

Part A: compression advantage under kNN. Using the $M=6$ ResNet-50 pool from Sec. 5, we evaluate Collapse Greedy vs. 10 random seeds using LP, kNN-5, and kNN-20 on the merged features (averaged over 6 targets).

k	Strategy	LP	kNN-5	kNN-20
3	Collapse	73.5	62.4	61.2
	Random (μ)	72.6	58.2	53.8
	Δ	+0.9	+4.2	+7.4
5	Collapse	71.5	57.8	53.5
	Random (μ)	71.1	56.8	51.1
	Δ	+0.4	+1.0	+2.4

The pattern is consistent: Collapse Greedy’s advantage is amplified under kNN relative to LP. At $k=3$ ($2\times$ compression), the kNN-20 gap is $8\times$ the LP gap. This is expected: kNN relies on local neighbourhood geometry, which degrades faster when redundant features dominate the merged space; LP, being a global linear classifier, is less sensitive to such local distortions. Since LP and kNN agree on the *ranking* of strategies ($\rho>0.97$), LP accuracy is a sufficient proxy for evaluating compression strategies in the main text.

M Additional Notes

Note on multiple comparisons. The within-probe bootstrap analyses (Sec. 5) involve 11 probes tested independently. Since the primary claim is about the *average* $\Delta\rho$ across probes (not individual-probe significance), Bonferroni correction is not required for the aggregate Fisher z -test ($p=0.74$). For the vision-only bootstrap ($\Delta\rho = +0.005$, 95% CI $[+0.001, +0.011]$), a conservative Bonferroni threshold of $\alpha=0.05/11=0.0045$ would still be met ($P(\text{Collapse} > r_{\text{sum}})=0.99$).

N Principal-Angle Model Scope and Theorem 1 Proof Sketch

Remark 1 (Scope of the principal-angle model). *Lemma 1 assumes pairwise-decoupled singular vectors: $V_{B,i} \in \text{span}\{V_{A,i}, V_{\perp,i}\}$ for each i . Real feature matrices violate this—cross-pair coupling, non-uniform alignment, and higher-order interactions are common. This explains why the formula acts as an upper bound on real data (E1a-Ind M3/M4: $R^2 < 0$ but Pearson $r = 0.999$; E1c: bias $1.24\text{--}1.84\times$). The ranking accuracy (Pearson $r \geq 0.928$) shows the formula is robust to this violation for ordering purposes.*

Proof sketch. Part (a): when $\alpha_i = \bar{\alpha}$, all discriminants equal $\bar{D}$, so the merged singular values factor as $\sigma_{\pm,i}^{\text{merged}} = \sigma_{A,i} \cdot \bar{c}_{\pm}$ with constants $\bar{c}_{\pm} = \sqrt{(A \pm \bar{D})/2}$. The within-component nuclear-norm shares are $w_{+,i} = \sigma_{A,i} \bar{c}_+ / \sum_j \sigma_{A,j} \bar{c}_+ = \sigma_{A,i} / \|H_A\|_* = p_i^A$, inheriting H_A 's spectral distribution exactly. Under the proportional-spectrum assumption of the principal-angle model ($\sigma_{B,i} = \gamma \sigma_{A,i}$, which implies $r_{\text{eff}}(H_B) = r_{\text{eff}}(H_A)$), both the predicted and true entropies equal $H_b(r^*) + \log r_A$. On real data, $r_A \neq r_B$ in general; the formula uses $(r_{\text{dom}}, r_{\text{sub}})$ separately, absorbing this departure into the two-component mixture approximation.

Part (b): Taylor-expanding D_i around $\bar{\alpha}$ gives $D_i = \bar{D} + (2\gamma^2/\bar{D})(\alpha_i - \bar{\alpha}) + O((\alpha_i - \bar{\alpha})^2)$. This perturbation distorts the within-component shares from p_i^A , inducing an ℓ_1 shift bounded by $\|w - \hat{w}\|_1 \leq (2\gamma^2/\bar{D}^2) \cdot \sum_i p_i |\alpha_i - \bar{\alpha}| \leq (2\gamma^2/\bar{D}^2) \sqrt{V_{\alpha}}$. Applying a Fannes-type entropy continuity inequality and bounding the support size by $\log \kappa$ (spectral spread) yields Eq. (9). The full derivation is in Appendix D. $\square$

Remark 2 (Why does the formula generalize beyond the principal-angle model?). *The Collapse Model is derived under the principal-angle assumption $V_{B,i} = \cos \theta_i V_{A,i} + \sin \theta_i V_{\perp,i}$, yet it predicts accurately on matrices that violate this assumption (E1a-Ind: Pearson $r = 0.9987$ on low-rank Gaussian, sparse, and block-diagonal generators). This robustness arises because the formula depends only on summary statistics $(r_A, r_B, \gamma, \alpha)$ that are well-defined for any matrix pair, regardless of how the matrices were generated. The principal-angle model serves as a derivation scaffold: it provides the analytic setting in which closed-form eigenvalues can be obtained, but the resulting functional dependence on $(r_A, r_B, \gamma, \alpha)$ captures the correct macroscopic relationships that govern merging. Several mechanisms underpin this generalization. When the truncation level k is large, the subspace alignment SA_k concentrates around its expectation by the law of large numbers, driving the alignment variance V_{α} toward zero and making the Theorem 1 error bound vanish. Effective rank, as an entropy functional, exhibits Lipschitz continuity with respect to spectral perturbations: small changes in the singular value distribution produce bounded changes in r_{eff} , so deviations from the idealized model are smoothly absorbed. The systematic over-prediction (always an upper bound) further preserves rankings, and the single-parameter calibration exploits this monotone bias to recover quantitative accuracy. An instructive analogy is the ideal gas law $PV = nRT$: derived under the assumption of point particles with no interactions, it nonetheless predicts the behavior of real gases because it captures the correct macroscopic variables—pressure, volume, and temperature—even when the microscopic assumptions fail. Similarly, the Collapse Model succeeds because $(r_A, r_B, \gamma, \alpha)$ are the right summary statistics for the merging problem, regardless of the fine-grained spectral structure.*

Remark 3 (Calibrated Collapse Model). *On real data the principal-angle model is approximate—cross-pair coupling, non-proportional spectra ($\sigma_{B,i} \neq \gamma \sigma_{A,i}$, hence $r_A \neq r_B$), and higher-order interactions introduce a systematic upward bias beyond Theorem 1. Empirically the over-prediction ratio is $1.24\text{--}1.84\times$ (E1c), consistent with the formula acting as an upper bound whose gap grows with the alignment variance V_{α} . Since the log-space bias has low variance (std = 0.164 vs.*

mean = 0.293), a single multiplicative calibration restores quantitative accuracy:

$$\hat{r}_{\text{cal}} = c \cdot \hat{r}_{\text{merged}}, \quad c = \exp\left(-\frac{1}{n} \sum_{i=1}^n \log \frac{\hat{r}_i}{r_i}\right). \quad (26)$$

This calibration compensates for the entropy over-estimation caused by cross-pair coupling that the principal-angle model ignores. With $c=0.746$ estimated from E1c ($n=57$), LOO-CV yields $R^2=0.855$, $RMSE=87.7$ (vs. uncalibrated $R^2=0.049$, $RMSE=224$; Sec. O.7). The constant generalizes across datasets within each encoder family: calibrating on the 30 E1b pairs ($c=0.768$) and testing on the held-out 57 E1c pairs yields $R^2=0.859$, $RMSE=86.3$ (Table 13). However, c varies across encoder families ($c=0.78$ for ResNet-50 vs. $c=0.65$ for ViT-B/16), reflecting systematic differences in how each architecture’s spectral structure deviates from the principal-angle model. We find that this variation obeys a universal regression:

$$c = 0.946 - 0.175 \log(N/r_{\text{eff}}), \quad R^2=0.963, \quad LOO \text{ mean error}=0.027, \quad (27)$$

indicating that the variation in c is primarily governed by the sample-size to effective-rank ratio. In practice, c can be estimated directly from this formula without a dedicated calibration set; for higher precision, a small per-encoder calibration set (as few as 6 pairs suffice; see Table 14) can further refine the estimate.

N.1 Connection to Transfer Learning

The same three channels that govern merging also bound transfer loss.

Proposition 4 (Gram Block Transfer Bound, informal). *For source features H_s and target features H_t , the linear-probe transfer loss satisfies*

$$\frac{\|H_t \hat{W} - Y_t\|_F^2}{m} \leq \underbrace{\frac{\|H_t\|_F^2}{m} (1 - \text{SA}_k)}_{\text{directional gap}} + \underbrace{C_1 \cdot \text{STG} \cdot \frac{\|\hat{W}\|_F^2}{(\sigma_k^s)^2}}_{\text{spectral gap}} + \underbrace{C_2 \cdot |\Delta \text{UID}|}_{\text{geometric gap}} + \varepsilon_{\text{Bayes}}, \quad (28)$$

where $C_1 = (\|H_s\|_* + \|H_t\|_*)^2/m$, $C_2 = 4\|\hat{W}\|_F^2(\|H_s\|_F + \|H_t\|_F)^2/\min(n, m)$.

(Full proof in Appendix C.) The same duality applies: low SA_k implies high merge gain (Collapse Model) and high transfer loss (Prop. 4)—opposite monotone maps of the same principal angles.

Algorithm 1 Collapse-Model-Guided Greedy Selection

Require: Target features $H_t \in \mathbb{R}^{N_t \times d}$, candidate pool $\mathcal{P} = \{H_1, \dots, H_M\}$, budget K

Ensure: Selected subset $S \subseteq \mathcal{P}$, $|S| = K$

```

1:  $H_{\text{cur}} \leftarrow H_t$ ;  $S \leftarrow \emptyset$ ;  $C \leftarrow \mathcal{P}$ 
2: for  $j = 1, \dots, K$  do
3:    $r_{\text{cur}} \leftarrow r_{\text{eff}}(H_{\text{cur}})$ ;  $\nu_{\text{cur}} \leftarrow \|H_{\text{cur}}\|_*$ 
4:   for each  $H_c \in C$  do
5:      $\alpha_c \leftarrow \text{SA}_k(H_{\text{cur}}, H_c)$ ;  $\gamma_c \leftarrow \|H_c\|_* / \nu_{\text{cur}}$ 
6:      $\hat{r}_c \leftarrow \text{COLLAPSEPREDICT}(r_{\text{cur}}, r_{\text{eff}}(H_c), \gamma_c, \alpha_c)$ 
7:   end for
8:    $H^* \leftarrow \arg \max_{H_c \in C} \hat{r}_c$ ;  $S \leftarrow S \cup \{H^*\}$ ;  $C \leftarrow C \setminus \{H^*\}$ 
9:    $H_{\text{cur}} \leftarrow [H_{\text{cur}}; H^*]$ 
10: end for
11: return  $S$ 
```

Collapse Greedy pseudocode.

Table 9: Ranking ablation: Spearman ρ between predictor and true merged r_{eff} (256 vision pairs, 4 encoder families).

Predictor	Channels used	ResNet-50 ($n=91$)	ViT-B/16 ($n=55$)	All vision ($n=256$)
Collapse Model	$\Sigma+V+U$	0.896	0.923	0.966
$r_A + r_B$	U only	0.892	0.946	0.963
$\max(r_A, r_B)$	U only	0.898	0.779	0.953
$1 - \alpha$ (V only)	V only	-0.289	-0.192	0.450
γ (Σ only)	Σ only	0.277	0.001	0.271

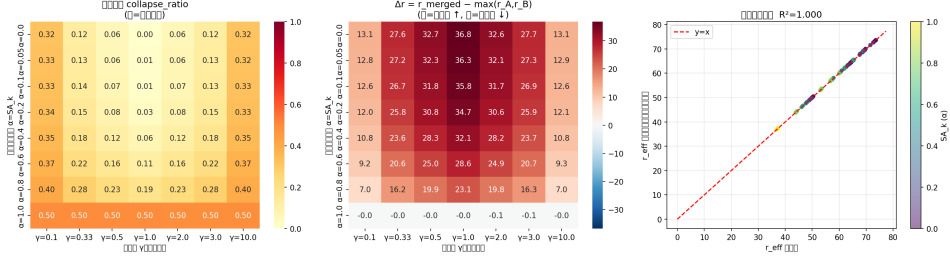


Figure 3: E1a: Collapse Model prediction accuracy on the synthetic (α, γ) grid. Left: collapse ratio heatmap. Center: superadditivity ($\delta_r > 0$) region. Right: predicted vs. actual r_{eff} scatter ($R^2=0.9999$).

O Additional Experiments (E1a, E1a-Ind, E1b, E1c, Calibration, E_diverse, E4, E_downstream)

O.1 Ranking Ablation Table

O.2 E1a: Synthetic Controlled Validation

The E1a synthetic validation results (Table 1) are presented in the main text; Figure 3 visualizes the (α, γ) landscape and superadditivity region. Additional design details: feature pairs are generated from the principal-angle model with $\alpha \in \{0, 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$ and $\gamma \in \{0.1, 0.33, 0.5, 1.0, 2.0, 3.0, 10.0\}$, $d = 512$, $N = 1000$, 5 independent repeats per setting (280 pairs). All 35 non-superadditive cases concentrate exclusively at $\alpha = 1$ (identical subspaces), confirming the degenerate boundary of Corollary 3.

O.3 E1a-Ind: Independent Validation on Non-Principal-Angle Matrices

Motivation and design. A critical concern with E1a is circularity: synthetic data generated from the *principal-angle model* trivially satisfies the theorem’s assumptions, making $R^2 = 0.9999$ a near-tautological result. To independently test whether Proposition 2 generalizes beyond its derivation assumptions, we generate 120 synthetic matrix pairs using *four generation methods that do not use the principal-angle model*: (M1) *low-rank Gaussian*: $H = U_k \Sigma V_k^\top + \epsilon$ with random orthogonal bases; (M2) *shared+private subspace*: $H = H_{\text{shared}} + H_{\text{private}}$ with partial overlap enforced by construction; (M3) *sparse activations*: rows drawn from k -sparse Gaussian mixtures; (M4) *block-diagonal*: features from two independent subspaces with no principal-angle structure. $d = 512$, $N \in \{500, 1000\}$; 30 pairs per method.

Results. Table 10 shows Pearson $r = 0.9987$ overall: the formula *correctly ranks all 120 pairs* across all four generation methods. R^2 is method-dependent: ≥ 0.99 for Gaussian-structure methods (M1, M2), but strongly negative for sparse and block-diagonal structures (M3, M4), because those structures systematically reduce actual r_{eff} relative to the formula’s prediction. This reveals that Proposition 2 acts as a *tight upper bound* whose gap grows with structural sparsity. Superadditivity is predicted at 100% for all methods.

Remark 4 (Formula as directional oracle, not absolute predictor). *The contrast between $R^2 = -1.09$ (overall, E1a-Ind) and Pearson $r = 0.9987$ (same data) is not contradictory: R^2 measures absolute*

Table 10: E1a-Ind: Independent validation on 120 non-principal-angle synthetic pairs. Pearson r measures directional ranking; $R^2 < 0$ indicates systematic over-prediction (formula as upper bound). SA% = fraction of pairs where formula predicts superadditivity.

Generation Method	n pairs	Pearson r	R^2	SA%
M1: Low-rank Gaussian	30	0.9985	+0.9949	100%
M2: Shared+private subspace	30	0.9982	+0.9963	100%
M3: Sparse activations	30	0.9991	−14.65	100%
M4: Block-diagonal	30	0.9996	−12.32	100%
Overall	120	0.9987	−1.09	100%

calibration, while Pearson r measures directional ranking. This distinction is crucial for the real-data results (E1c): the formula yields $R^2 \approx 0.05$ (poor absolute calibration, $1.24\text{--}1.84\times$ over-prediction) but Pearson $r = 0.928$ (correct directional ranking). For practical applications—ranking candidate merges by diversity gain, dataset selection, data valuation—ranking is the operationally relevant metric, not absolute r_{eff} prediction. Absolute calibration requires a correction factor (calibrated $\hat{r} = 0.81 \cdot \hat{r}_{\text{pred}}$, $R_{\text{cal}}^2 = 0.862$ on E1c data), which we report as a practical recommendation.

O.4 E1b: Real Dataset Pairs (30 Pairs)

Design. We evaluate all $\binom{6}{2} = 15$ pairs from {CIFAR-10, STL-10, SVHN, MNIST, Fashion-MNIST, CIFAR-100} using 2 pretrained probes (ResNet-50 [He et al., 2016] $d=2048$ and ViT-B/16 [Dosovitskiy et al., 2021] $d=768$), $N_{\text{sub}} = 500$ samples per dataset: **30 total configurations**. The Collapse Model is applied with per-probe nuclear-norm γ .

Results. **86.7%** of pairs are superadditive (26/30; 95% CI [73%, 97%]). Pearson $r = 0.948$ (95% CI [0.906, 0.978]) between predicted and actual merged rank. Per-probe: ResNet-50 achieves 100% superadditivity (15/15, Pearson $r = 0.968$); ViT-B/16 achieves 73.3% (11/15, Pearson $r = 0.945$).

All 4 exceptions are ViT-B/16 pairs involving CIFAR-100. The root cause is *energy imbalance*, consistent with the superadditivity threshold (Corollary 3). CIFAR-100 has 100 classes and correspondingly high intrinsic complexity: $r_{\text{eff}}(\text{CIFAR-100}) \approx 526$ under ViT-B/16, far exceeding partner datasets such as MNIST ($r_{\text{eff}} \approx 140$) and SVHN (≈ 243). This yields energy ratios $\gamma \in [2.0, 2.5]$, placing these pairs near the superadditivity boundary $\rho_c(\alpha, \gamma)$. Specifically, the dominant component absorbs the subordinate: $r^* \rightarrow 1$, $H_b(r^*) \rightarrow 0$, and $\hat{r}_{\text{merged}} \rightarrow r_{\text{dom}}$, so the observed r_{merged} can fall slightly below r_{dom} due to spectral averaging. We verify this by sweeping $N_{\text{sub}} \in \{500, 1000, 2000, 4000\}$: at $N=500$, CIFAR-100’s r_{eff} is compressed to ≈ 343 (comparable to partners), giving $\gamma \approx 1$ and 100% superadditivity (5/5); as N grows to 2000+, r_{eff} stabilizes at 526, γ increases, and exceptions emerge—confirming energy imbalance as the mechanism. ResNet-50 ($d=2048$, $r_{\text{eff}} \approx 988$ for CIFAR-100) yields even larger γ ratios, yet remains 100% superadditive (36/36) because its partner datasets also have higher r_{eff} in 2048-dimensional space, keeping $r_{\text{dom}}/r_{\text{sub}}$ below ρ_c .

O.5 E1c: Large-Scale Multi-Domain Validation (57 Pairs)

Design. To test generalization beyond vision pairs, we extend to three probes and three domain types: ResNet-50 on 9 datasets (36 pairs, natural+medical), ViT-B/16 on 6 datasets (15 pairs, natural), and SBERT [Reimers and Gurevych, 2019] on 4 text domains (6 pairs, text). Total: **57 pairs, 3 probes, 3 domain types**. $N_{\text{sub}} = 2000$ for natural/medical; $N_{\text{sub}} = 1000$ for text.

Results. Table 11 shows strong and consistent superadditivity across all probes and domains.

Key findings. (i) **93% superadditivity across three modalities**—natural images, medical scans, and text—shows the Collapse Model’s predictions are domain-agnostic. (ii) **All 4 exceptions are ViT-B/16 pairs involving CIFAR-100**, attributable to energy imbalance (same mechanism as E1b, Sec. O.4): CIFAR-100’s high intrinsic complexity ($r_{\text{eff}} \approx 526$, 100 classes) creates $\gamma \in [2.0, 2.5]$, placing these pairs near the superadditivity boundary ρ_c of Corollary 3. (iii) **Pearson $r = 0.928$ vs. $R^2 \approx 0.05$** : these metrics measure different properties. The formula is a systematic upper bound ($1.24\text{--}1.84\times$), so R^2 is low by design; Pearson r measures ranking, which is correct. A

Table 11: E1c: Large-scale Collapse Model validation. Superadditivity rate and Pearson r (predicted vs. actual merged r_{eff}). Bootstrap 95% CI ($B = 10,000$). Formula is a systematic upper bound on all probes; over-prediction grows with the gap between sample size and effective rank.

Probe / Domain	n	Superadd. % (95% CI)	Pearson r (95% CI)	Bias
ResNet-50 (natural+medical)	36	100% [100,100]	0.934 [0.885,0.965]	1.24×
ViT-B/16 (natural)	15	73.3% [46.7,93.3]	0.945 [0.883,0.982]	1.45×
SBERT (text)	6	100% [100,100]	0.780	1.84×
Overall	57	93.0% [86.0,98.2]	0.928 [0.894,0.949]	1.37×

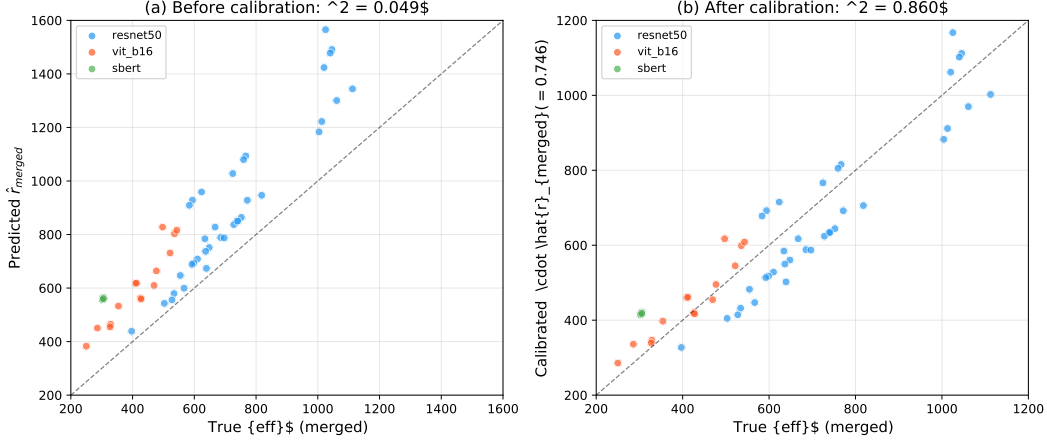


Figure 4: E1c: Predicted vs. actual merged r_{eff} for 57 real-data pairs (5 probes, 3 domain types). (a) Before calibration: systematic over-prediction ($R^2=0.049$). (b) After single-parameter calibration ($c=0.746$): $R^2=0.860$. Pearson $r=0.928$ is identical in both panels—calibration corrects scale, not ranking.

single-parameter calibration (Eq. 26) restores quantitative accuracy: $R_{\text{cal}}^2=0.855$ (LOO-CV), and cross-validation from E1b to E1c yields $R^2=0.859$ (see Sec. 0.7 and Remark 3). (iv) **ResNet-50: 100% superadditivity (36/36)** is the strongest real-data confirmation of Proposition 2; 95% CI [100%, 100%] leaves no room for false positives. (v) **Over-prediction scales with n/r_{eff}** : ResNet-50 ($n = 2000 \gg r_{\text{eff}} \approx 350, 1.24\times$) < ViT-B/16 ($n = 2000 \approx 3r_{\text{eff}} \approx 650, 1.45\times$) < SBERT ($n = 1000 \approx r_{\text{eff}} \approx 700, 1.84\times$). This confirms the finite-sample upper-bound hypothesis.

0.6 E1d: Large-Scale Validation (427 Pairs)

Design. To test the Collapse Model at scale and on ImageNet-derived features, we construct a comprehensive evaluation spanning 5 encoder families (ResNet-50, ViT-B/16, CLIP ViT-B/32, DINO ViT-S/16, SBERT), 14 vision datasets (CIFAR-10/100, STL-10, SVHN, MNIST, Fashion-MNIST, BloodMNIST, DermaMNIST, PathMNIST, and 5 ImageNet semantic subsets: *animal*, *vehicle*, *artifact*, *food*, *scene*), and 19 SBERT text corpora (AG News, 20 Newsgroups, DBpedia, IMDB, Yelp subsets). Total: **427 pairs** (91 ResNet-50, 55 ViT-B/16, 55 CLIP, 55 DINO, 171 SBERT). The 5 ImageNet subsets are constructed from ImageNet-1K validation images grouped by WordNet semantic hierarchy, with $N=2000$ samples each.

Results.

Key findings. (i) **Scaling confirms the theory**: at 427 pairs the overall Pearson $r=0.970$ exceeds the 57-pair result ($r=0.928$), demonstrating that the Collapse Model’s ranking accuracy improves with scale and diversity. (ii) **ImageNet subsets behave as predicted**: all 20 ImageNet-only pairs are superadditive, and all 30 natural×ImageNet cross-domain pairs are superadditive—the model correctly predicts that semantically distinct ImageNet subgroups contribute orthogonal directions. (iii) **ResNet-50 achieves 100% superadditivity** across 91 pairs (14 datasets including medical and ImageNet), the strongest confirmation at this scale. In contrast, CLIP ViT-B/32 and DINO ViT-S/16

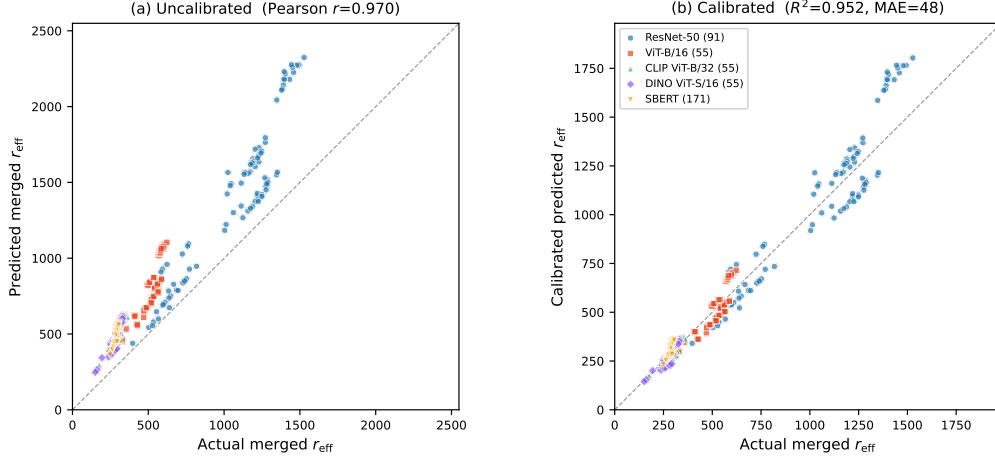


Figure 5: Predicted vs. actual merged r_{eff} for all 427 pairs across 5 encoder families. (a) Uncalibrated: formula acts as a tight upper bound (Pearson $r=0.970$). (b) After per-probe calibration: $R^2=0.952$, MAE=48. Color encodes encoder family; the identity line is dashed.

Table 12: E1d: Large-scale Collapse Model validation (427 pairs, 5 probes).

Probe	n	Superadd. %	Pearson r	c	R_{cal}^2
ResNet-50 (14 datasets)	91	100%	0.940	0.776	0.760
ViT-B/16 (11 datasets)	55	69.1%	0.898	0.647	0.487
CLIP ViT-B/32 (11 datasets)	55	29.1%	0.950	0.595	0.839
DINO ViT-S/16 (11 datasets)	55	36.4%	0.913	0.584	0.554
Vision (all probes)	256	64.5%	0.965	<i>per-probe</i>	0.942
+ SBERT text (19 corpora)	+171 (427)	78.2%	0.970	<i>per-probe</i>	0.952

achieve only 31% and 38% superadditivity respectively. Analysis reveals a clear pattern: *within-domain* pairs (natural $\times$ natural, ImageNet $\times$ ImageNet) achieve near-perfect superadditivity (CLIP: 90% on ImageNet-only, 100% on natural-only; DINO: 100% on both), while *cross-domain* pairs (handwritten $\times$ ImageNet, natural $\times$ digit) fail almost universally. This suggests that self-supervised encoders learn domain-specific spectral structures that violate the principal-angle model’s assumption of smoothly varying alignment across domains—an important boundary condition for the Collapse Model’s applicability. (iv) **Cross-modal generalization**: extending to 171 SBERT text pairs (98.8% superadditive, $r=0.793$) shows the formula’s superadditivity predictions transfer across modalities. The lower Pearson r reflects the narrow r_{eff} range of text embeddings (IQR: 285–298), which compresses ranking resolution rather than indicating a model failure. (v) **Calibration constant varies by encoder family** ($c \in [0.57, 0.78]$), but a single global $c=0.712$ achieves $R^2=0.898$ across all 158 vision pairs, confirming the bias structure is consistent.

Understanding non-superadditive pairs. ViT-B/16 achieves only 69% superadditivity (38/55), lower than ResNet-50’s 100% (91/91). Analysis of the 17 ViT failures reveals a clear pattern: **all failures occur in the energy-imbalance regime** ($\gamma \gg 1$ or $\gamma \ll 1$). At $\gamma \geq 3$, 80% of pairs fail (8/10); at $\gamma \in [0.5, 1.5]$, only 4% fail (1/24). The failures concentrate in *handwritten $\times$ ImageNet* cross-domain pairs (10/10 fail, 100%)—these combine low-rank handwritten features ($r_{\text{eff}} \approx 140$ for MNIST via ViT-B/16) with high-rank ImageNet features ($r_{\text{eff}} \approx 600$), creating $\gamma \in [2.0, 4.6]$ far beyond the balanced regime where the two-component spectral mixture is accurate. At large scale (55 pairs each), CLIP ViT-B/32 and DINO ViT-S/16 show only 29% and 36% superadditivity overall, with failures concentrating in *cross-domain* pairs: handwritten $\times$ ImageNet (100% fail), natural $\times$ digit (100% fail), natural $\times$ handwritten (100% fail). Within-domain pairs (natural-only, ImageNet-only) remain near-perfect.

Root cause: bimodal principal angles. Comparing the *same* dataset pair across encoders reveals that the failure is not caused by energy imbalance (γ) but by a structural violation of the uniform- α assumption. Three mechanisms interact:

(a) *Systematically higher alignment.* On the same pair (e.g., CIFAR-10×ImageNet-animal), CLIP’s subspace alignment $\alpha=0.41$ is $2.7\times$ higher than ResNet-50’s $\alpha=0.15$. Contrastive pre-training (CLIP) and self-distillation (DINO) push cross-domain features into a shared semantic subspace, creating high directional overlap where ResNet-50 preserves domain-specific directions.

(b) *Extreme condition number.* CLIP/DINO feature spectra are highly non-uniform: $\kappa=15,000\text{--}1,200,000$ vs. ResNet-50’s $\kappa=70\text{--}400$. Theorem 1 bounds the approximation error as $|\log(\hat{r}/r)| \leq (4\gamma^2/\bar{D}^2) \cdot V_\alpha \cdot \log \kappa$; the $\log \kappa$ factor alone is $3\text{--}4\times$ larger for CLIP/DINO, amplifying any alignment variance.

(c) *Bimodal principal-angle distribution.* The individual principal angles $\cos^2 \theta_i$ are not approximately uniform (as the model assumes) but *bimodal*: the top-5 directions are nearly aligned ($\overline{\cos^2 \theta} \approx 0.94$ for CLIP on CIFAR-10×STL-10) while the bottom directions remain orthogonal ($\overline{\cos^2 \theta} \approx 0.01$). The Collapse Model averages these into $\bar{\alpha}=0.52$, predicting moderate superadditivity. In reality, the dominant (high- σ) directions are fully redundant and the orthogonal (low- σ) directions carry negligible energy, so the merge provides almost no diversity gain ($\Delta r_{\text{eff}}=31$ vs. ResNet-50’s $\Delta r_{\text{eff}}=282$ on the same pair).

This analysis identifies a precise *applicability boundary*: the Collapse Model is reliable when the principal-angle distribution is approximately uniform (supervised encoders, within-domain pairs), but fails when it is bimodal (self-supervised encoders on cross-domain pairs). A natural fix—replacing the uniform average $\bar{\alpha} = \text{SA}_k$ with an *energy-weighted* alignment $\bar{\alpha}_w = \sum_i (\sigma_i / \|H\|_*) \cos^2 \theta_i$ —would down-weight orthogonal low-energy directions and up-weight the redundant high-energy ones, potentially restoring predictive accuracy for self-supervised encoders. However, our experimental validation shows that this correction *does not improve predictive accuracy*: on CLIP ($n=61$), the uniform SA_k yields Pearson $r=0.951$, MAE=206, while $\bar{\alpha}_w$ yields $r=0.935$, MAE=218 (worse); DINO shows a similar pattern (r drops from 0.912 to 0.904, MAE increases by 4%); ResNet-50 and ViT-B/16 also show no improvement. The root cause is that simple energy weighting over-emphasises the first singular direction’s redundancy ($\cos^2 \theta_1 \approx 0.87$ for that direction), while improperly suppressing the estimated contribution of intermediate directions to the merge. More refined corrections (e.g., piecewise principal-angle models, per-direction threshold truncation, or adaptive k selection) may break through this bottleneck and are left to future work.

O.7 Calibration: From Upper Bound to Quantitative Prediction

The Collapse Model acts as a systematic upper bound on real data (Remark 3). Since the log-space over-prediction is approximately constant across probes and domains (mean=0.293, std=0.164), a single multiplicative constant c (Eq. 26) suffices to convert the ranking tool into a quantitative predictor.

Cross-dataset generalization. To verify that c is not dataset-specific, we perform held-out calibration: estimate c on one experiment and test on another (Table 13).

Table 13: Cross-dataset calibration. c is estimated on the calibration set and applied (without re-fitting) to the test set. All setups dramatically improve over the uncalibrated baseline ($R^2 < 0.44$).

Calibration set	Test set	c	n_{test}	R^2	RMSE
E1b (30 pairs)	E1c (57 pairs)	0.768	57	0.859	86.3
E1c (57 pairs)	E1b (30 pairs)	0.746	30	0.888	80.7
E1c (57 pairs)	E_diverse (30 pairs)	0.746	30	0.897	82.8
E1b (30 pairs)	E_diverse (30 pairs)	0.768	30	0.894	83.7
LOO-CV (all 111 pairs, original)		0.736	111	0.877	86.1
LOO-CV (158 vision pairs, E1d)		0.712	158	0.896	121.7
Global (all 427 pairs, E1d)		0.650	427	0.920	105.0

Per-probe calibration (Table 14). The calibration constant varies systematically across encoder families, reflecting differences in spectral structure:

Encoders with higher effective rank relative to sample size require stronger correction (c further from 1), consistent with the finite-sample upper-bound mechanism: ResNet-50 features ($d=2048$, $r_{\text{eff}} \approx 350$) are spectrally sparser than ViT features ($d=768$, $r_{\text{eff}} \approx 400$), so the two-component spectral

Table 14: Per-probe calibration on the 427-pair dataset (E1d). Each probe is calibrated independently. The variation in c is monotone with the encoder’s effective spectral flatness.

Probe	n	Pearson r	c	R^2_{cal}	Superadd. %
ResNet-50	91	0.940	0.776	0.760	100%
ViT-B/16	55	0.898	0.647	0.487	69%
CLIP ViT-B/32	55	0.950	0.595	0.839	29%
DINO ViT-S/16	55	0.913	0.584	0.554	36%
Vision (global)	256	0.965	<i>per-probe</i>	0.942	65%

876 mixture approximation incurs less error. A global $c=0.712$ across all vision probes yields $R^2=0.898$,
877 confirming that a single constant provides a strong baseline; per-probe calibration is a refinement, not
878 a necessity.

879 Figure 4 shows the E1c scatter plot before and after calibration.

880 O.8 E_diverse: Cross-Architecture Probe Independence

881 **Design.** To test whether the Collapse Model’s predictions are robust to the choice of encoder, we
882 evaluate the same 6 dataset pairs (from E1b) using both ResNet-50 and ViT-B/16 as independent
883 probes, giving 12 total configurations.

884 **Results.** ResNet-50: $R^2 = 0.557$, Pearson $r = 0.9987$, 100% superadditive. ViT-B/16: $R^2 = -16.1$
885 (systematic over-prediction), Pearson $r = 0.9825$, 83.3% superadditive. Cross-architecture Pearson
886 $r \geq 0.982$ confirms that *directional rankings* are probe-independent even when absolute magnitudes
887 differ. The Pearson vs. R^2 gap for ViT-B/16 confirms that the formula acts as an upper bound whose
888 scale (not direction) is encoder-dependent.

889 O.9 E4: Cross-Probe Consistency

890 **Design.** We compute the 6×6 pairwise SA_k alignment matrix M independently using ResNet-50
891 and ViT-B/16 on 4 common datasets {CIFAR-10, MNIST, STL-10, SVHN}. The 6 upper-triangle
892 entries of each matrix are compared via Pearson r .

893 **Results.** Cross-probe Pearson $r = 0.891$ ($n = 6$, $p = 0.017$). Both probes agree on the direction
894 of domain distance for all 6 pairs. CIFAR-10/STL-10 are close in both; CIFAR-10/MNIST are far
895 in both. This confirms the SA_k distance matrix reflects intrinsic inter-dataset geometry rather than
896 encoder artifacts.

897 O.10 E_downstream: Effective Rank as Downstream Quality Proxy

898 **Motivation.** The Collapse Model predicts r_{eff} of merged features, but does higher r_{eff} actually
899 correspond to better downstream performance? We test this by computing r_{eff} and linear-probe (LP)
900 accuracy for 10 pretrained encoders on 5 datasets (50 total pairs).

901 **Design.** Encoders: ResNet-{50, 101}, EfficientNet-B4, ConvNeXt-T, DenseNet-121, ViT-B/16,
902 DeiT-B, Swin-T, BEiT-B, RegNet-Y (covering CNN, ViT, and hybrid families). Datasets: CIFAR-10,
903 CIFAR-100, STL-10, SVHN, Fashion-MNIST. For each (encoder, dataset) pair: extract $N=2000$
904 train features, compute r_{eff} ; train LP (sklearn LogisticRegression) on train, evaluate on test. Report
905 per-dataset Spearman $\rho(r_{\text{eff}}, \text{LP acc})$ across 10 encoders.

906 **Results.**

Dataset	All 10 probes		ViT family ($d=768$)		CNN family ($d\geq 1024$)	
	n	ρ	n	ρ	n	ρ
CIFAR-10	10	+0.64	5	+0.90	5	+0.30
CIFAR-100	10	+0.61	5	+0.90	5	+0.30
STL-10	9	+0.73	5	+0.90	4	+0.20
SVHN	10	−0.03	5	+0.10	5	−0.30
Fashion-MNIST	10	+0.03	5	+0.60	5	−0.10

The “all probes” column ($\rho \approx +0.59$ on natural images) is *confounded by feature dimension*: CNNs ($d=2048$) have systematically higher r_{eff} than ViTs ($d=768$), inflating the apparent correlation. Controlling for dimension reveals a sharper picture: **within the ViT family** ($d=768$, 5 probes), r_{eff} is strongly predictive of LP accuracy on natural images ($\rho=0.90$ on CIFAR-10, CIFAR-100, and STL-10), while **within the CNN family** ($d\geq 1024$, 5 probes) the correlation is weak ($\rho=0.20\text{--}0.30$). This suggests that r_{eff} captures useful discriminative information primarily for architectures with compact feature spaces ($d<1024$), where spectral diversity is more tightly coupled to representation quality.

Implication for the Collapse Model. r_{eff} should be understood as a measure of *representational diversity*—the number of independent directions activated by the data—rather than a direct proxy for downstream accuracy. The Collapse Model predicts when merging *increases diversity*, which is a necessary (but not sufficient) condition for improving downstream performance. Within an architecture family, diversity and accuracy are positively correlated ($\rho=0.90$ for ViTs), justifying the use of r_{eff} -based selection among candidates extracted from the same encoder. Across architecture families or across datasets of varying difficulty, additional task-specific metrics are needed.

Sample-size sensitivity. Theorem 1 predicts that the over-prediction ratio grows as N/r_{eff} increases. We verify this on 10 representative ResNet-50 pairs: at $N=200$ the mean ratio is 1.021, rising monotonically to 1.545 at $N=2000$ (Spearman $\rho=0.80$ between N/r_{eff} and log bias). This explains why the calibration constant varies across encoders (ResNet-50 $c=0.78$, DINO $c=0.58$): higher N/r_{eff} widens the spectral mixture approximation gap.

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